Week 6: Trees

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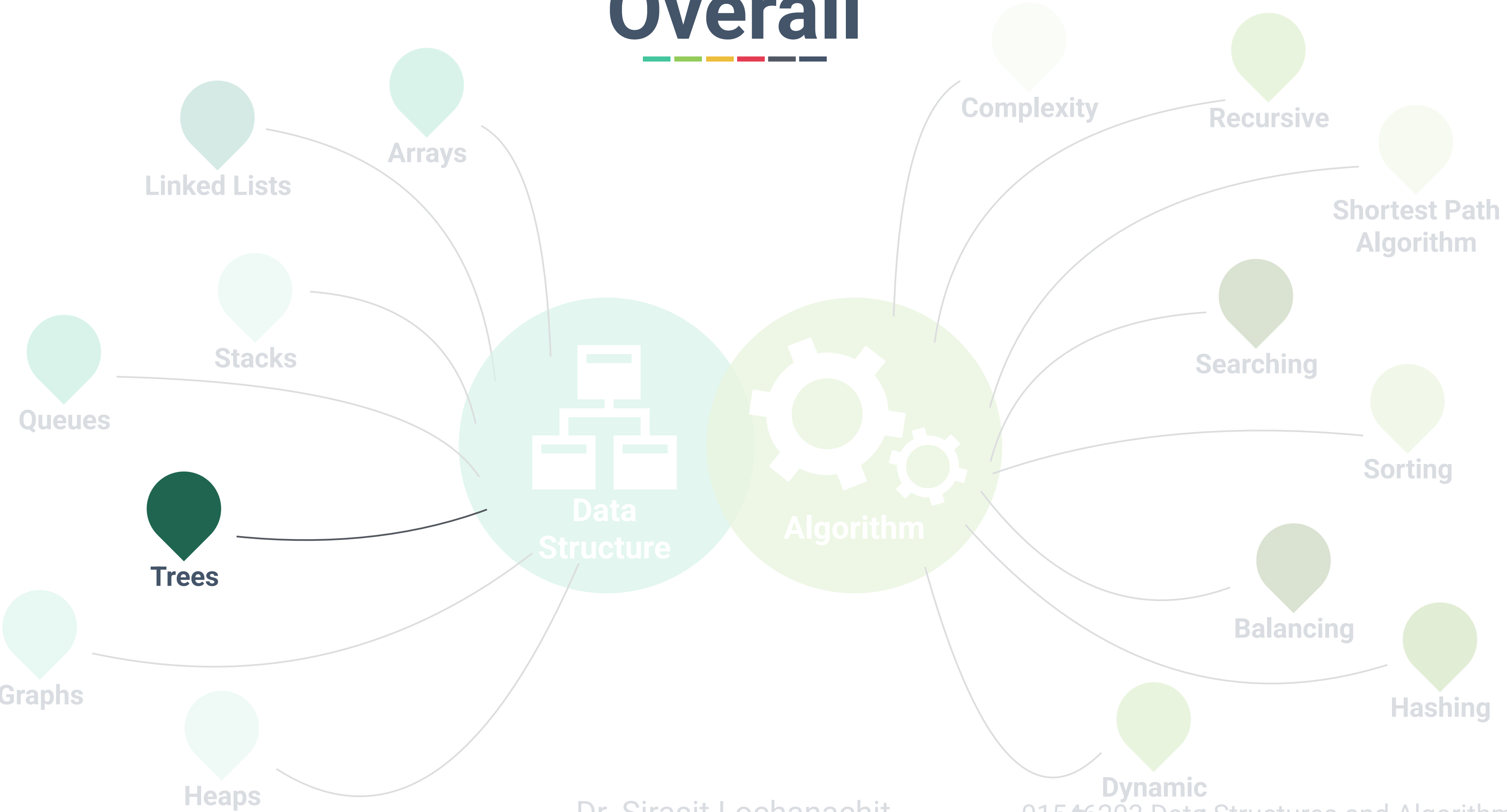
Overall


Arrays
Linked Lists
Stacks
Queues
Trees
Graphs
Heaps
Data Structure
Algorithm
Complexity
Recursive
Shortest Path Algorithm
Searching
Sorting
Balancing
Hashing
Dynamic Programming

Abstract Data Type

Linear:

- Arrays, Stacks, Queues, and Linked Lists
- Operations: push, pop, enqueue, dequeue
- Algorithms: Searching and sorting, insertion, deletion

Non-Linear:

- Trees and Graphs
- Algorithms: Traversal, Insertion and Deletion, Balancing, and etc.

Today's Outline

General Trees:

- Definition and examples
- Elements of Tree Structure

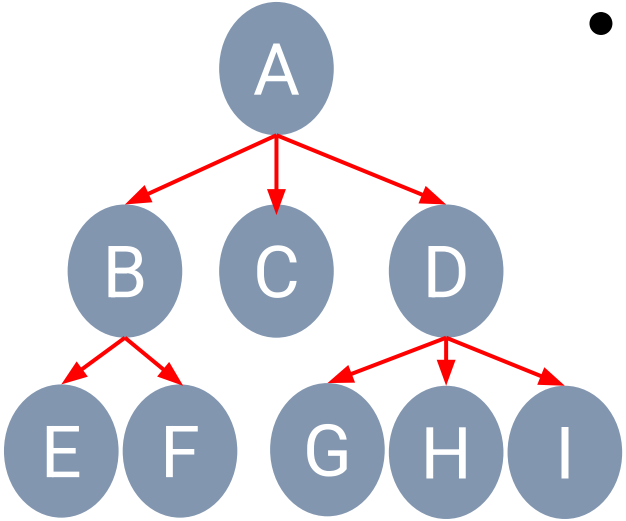
Binary Trees:

- Definition, properties, and types

Tree traversal algorithms:

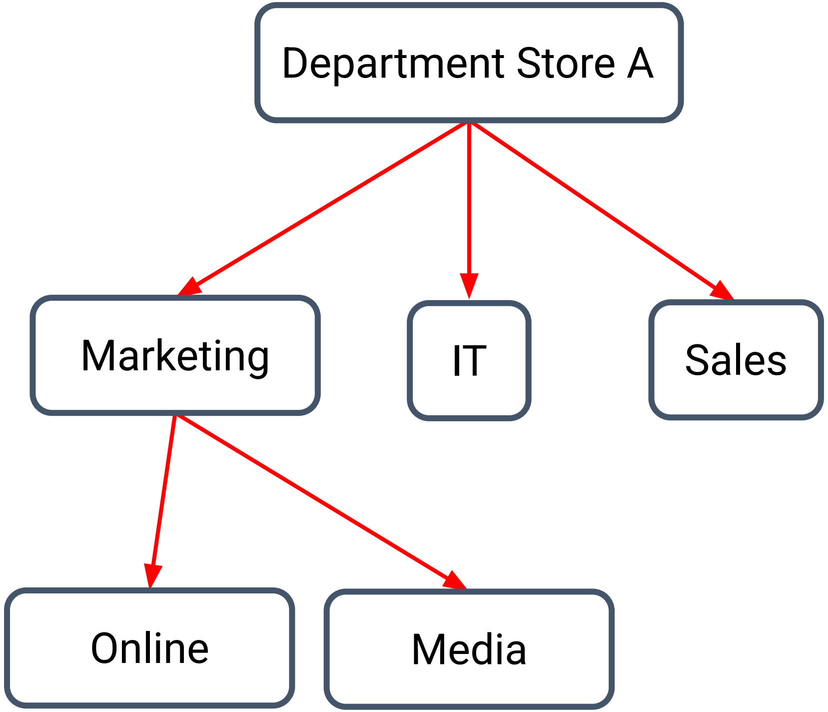
- Depth-first Traversal (Preorder, postorder, and inorder)
- Breadth-first Traversal

What is a Tree?



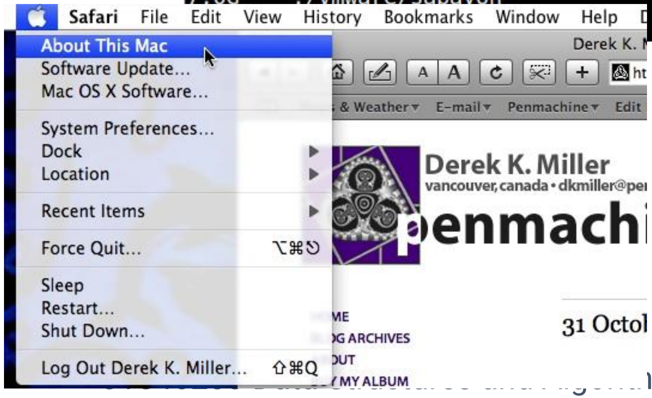
- A **tree** stores elements in a hierarchical structure.

Tree Example

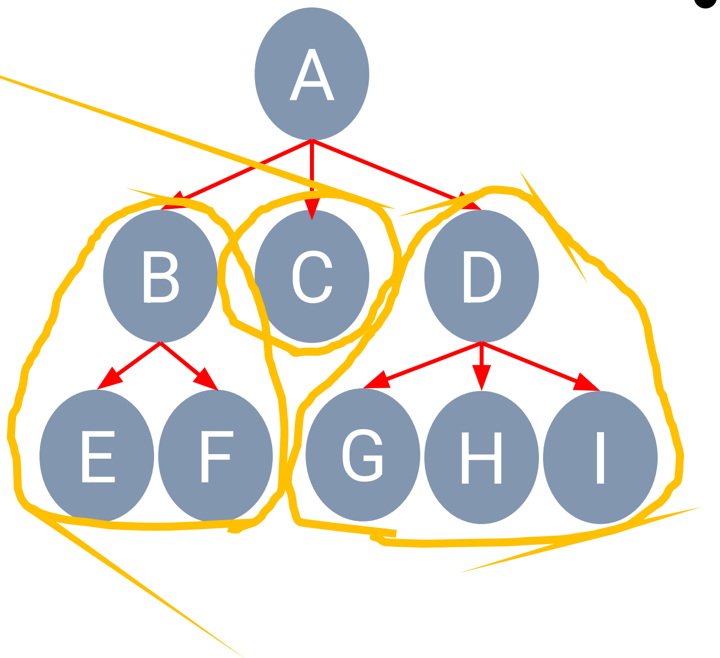


Tree Applications

- Tree represents natural organisation for data.
- Tree structure has been used widely in
  - File systems (Directory)
  - Graphical User Interfaces
  - Databases (Sub-categories, etc.)
  - Websites



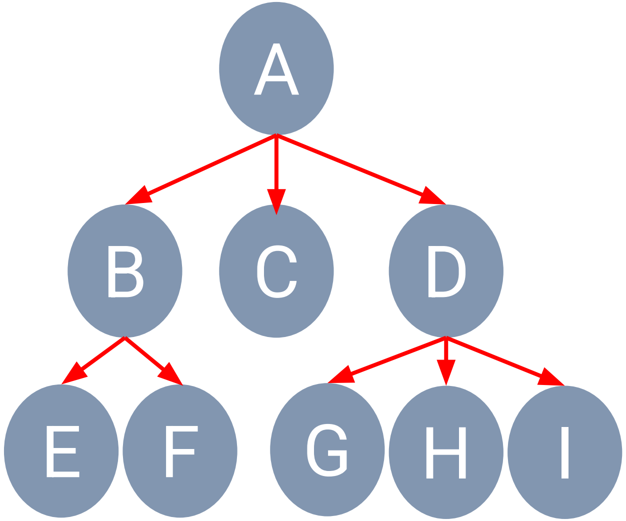
Formal Definition



- A **tree** is a collection of nodes that store elements with a **parent-child** relationship.

[1] Michael T. Goodrich et al., Data Structures and Algorithms in Python, 2013

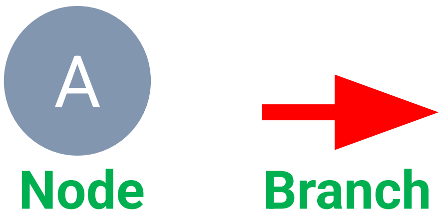
Formal Definition

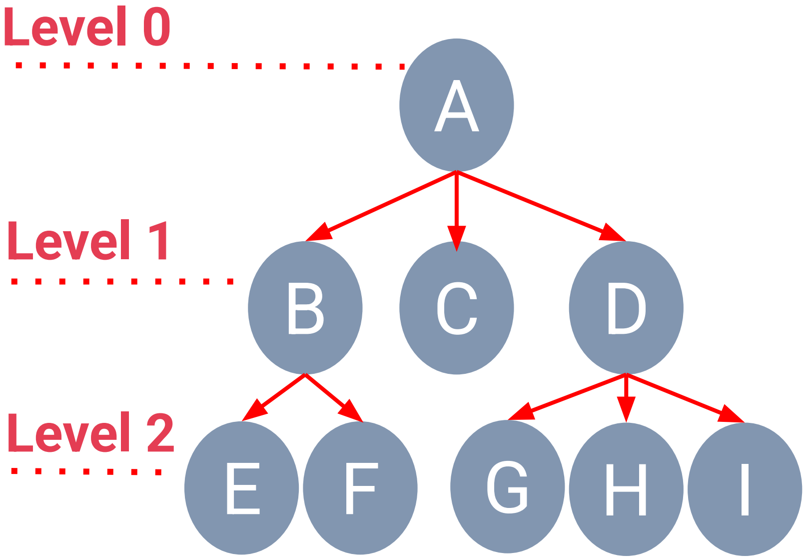


- **Root**
- **Edge**
- **Child**
- **Siblings**
- **Internal** or **<u>parent</u>** node
- **External** or **<u>leaf</u>** node
- **Ancestor**
- **Descendant**

[1] Michael T. Goodrich et al., Data Structures and Algorithms in Python, 2013

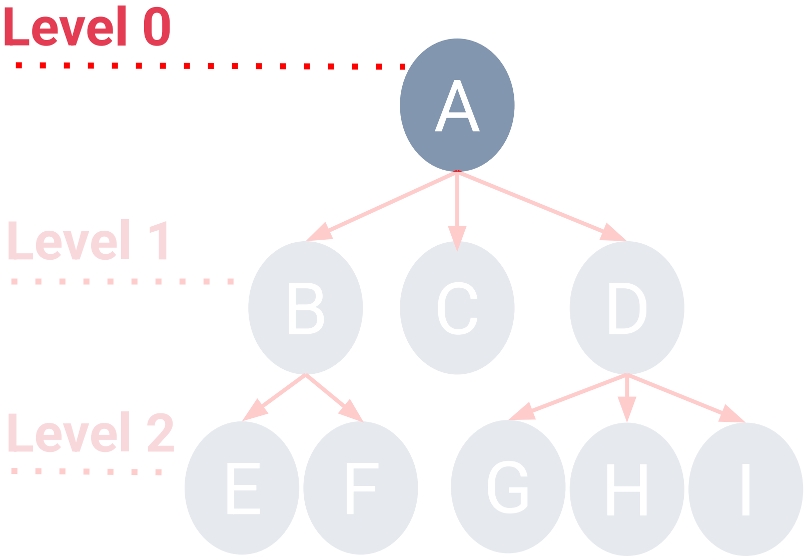
Basic Elements of Tree Structure





Basic Elements of Tree Structure

Root node is the *top element* of a tree.

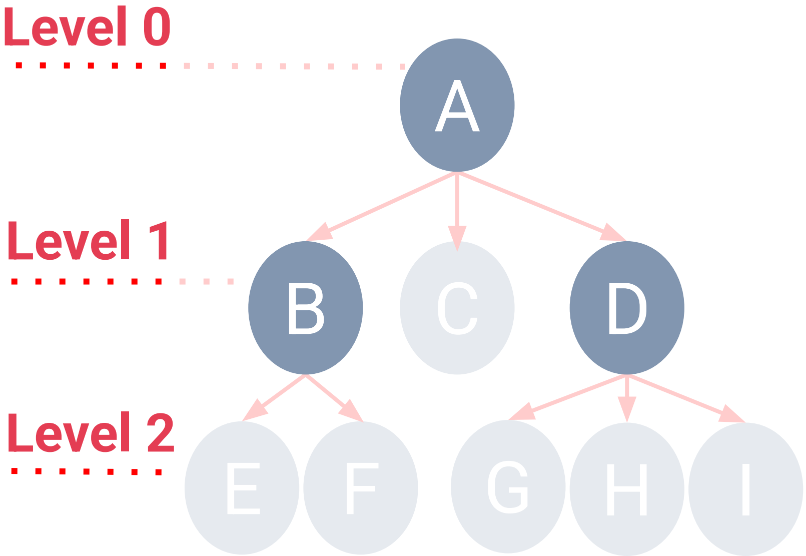


- ***Root*** Node: *A*

Basic Elements of Tree Structure

Parent node is an <u>internal</u> node that has one or more children nodes.

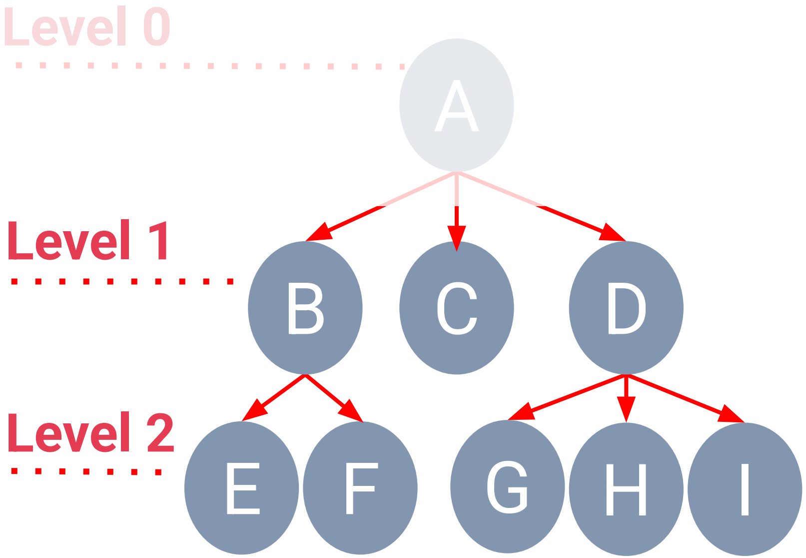
- ***Root*** Node: *A*
- ***Parents***: *A*, *B*, and *D*



Basic Elements of Tree Structure

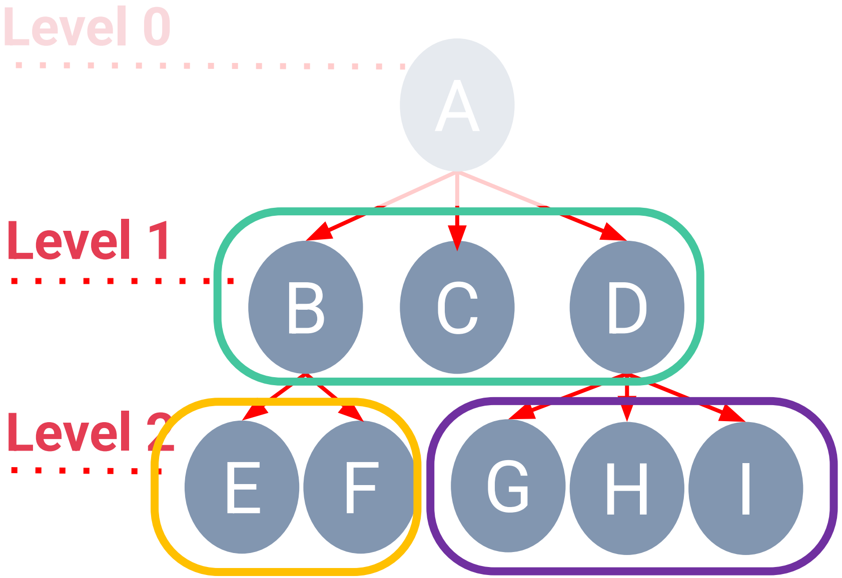
Children node is a node that has parent node.

- ***Root*** Node: *A*
- ***Parents***: *A*, *B*, and *D*
- ***Children***: *B*, *E*, *F*, *C*, *D*, *G*, *H* and *I*



Basic Elements of Tree Structure

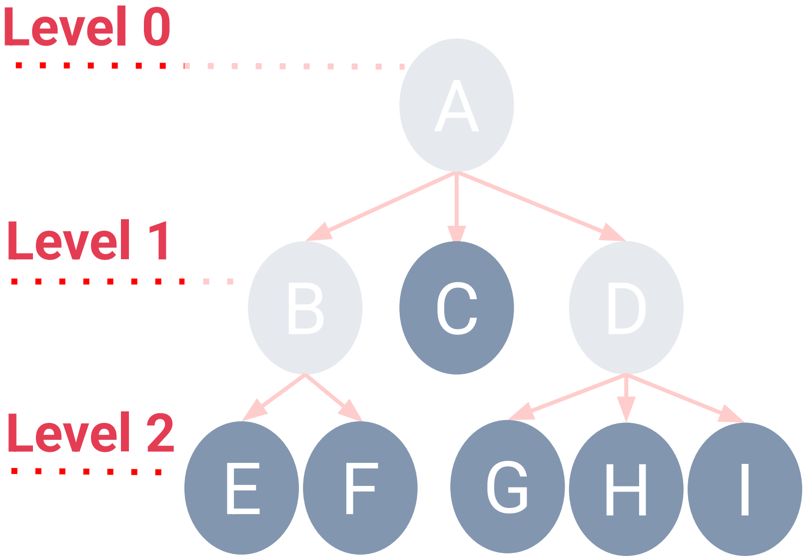
Two or more children nodes are ***siblings*** when they have the same parent node.



- ***Root*** Node: *A*
- ***Parents***: *A*, *B*, and *D*
- ***Children***: *B*, *E*, *F*, *C*, *D*, *G*, *H* and *I*
- ***Siblings***: {*B*, *C*, *D*}, {*E*, *F*}, {*G*, *H*, *I*}

Basic Elements of Tree Structure

Leaf **node is a node without any children.**

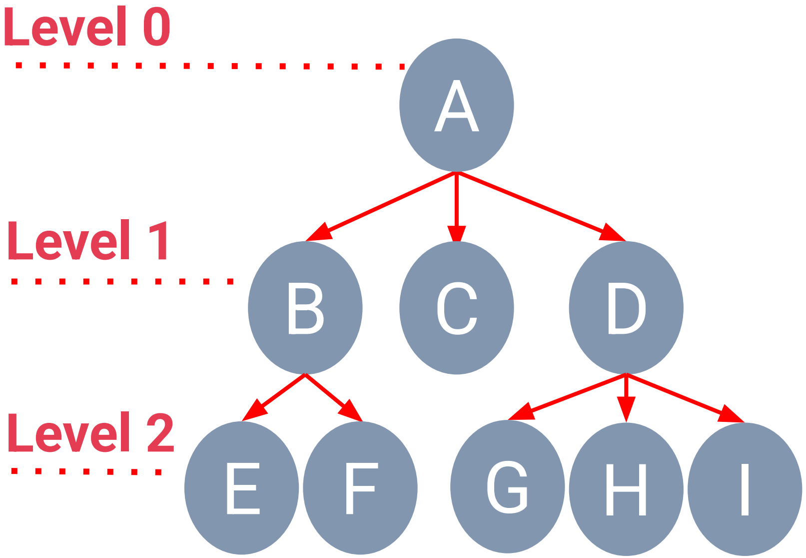


- ***Root*** Node: *A*
- ***Parents***: *A*, *B*, and *D*
- ***Children***: *B*, *E*, *F*, *C*, *D*, *G*, *H* and *I*
- ***Siblings***: {*B*, *C*, *D*}, {*E*, *F*}, {*G*, *H*, *I*}
- ***Leaf*** Nodes: *C*, *E*, *F*, *G*, *H* and *I*

Basic Elements of Tree Structure

A ***Degree of tree*** is a number of subtrees or edges/links.

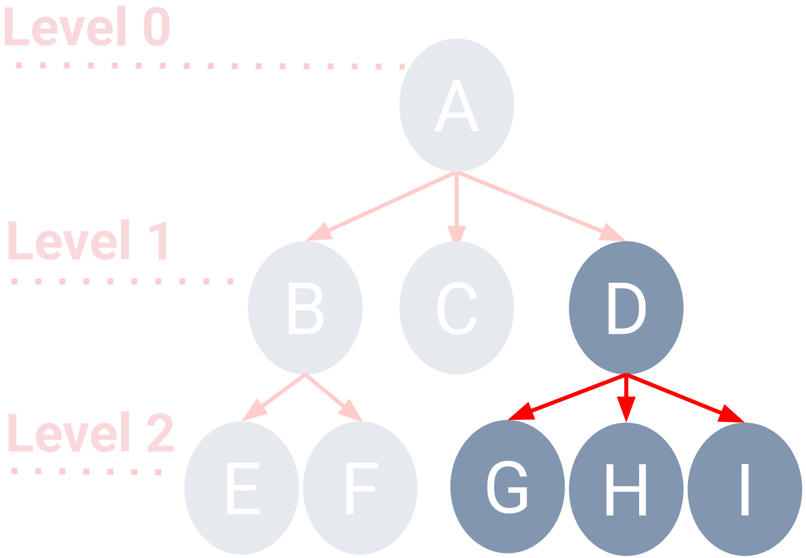
Edge is a pair between parent and child node.



- ***Children***: *B*, *E*, *F*, *C*, *D*, *G*, *H* and *I*
- ***Siblings***: {*B*, *C*, *D*}, {*E*, *F*}, {*G*, *H*, *I*}
- ***Leaf*** Nodes: *C*, *E*, *F*, *G*, *H* and *I*
- Degree of Tree is 8

Basic Elements of Tree Structure

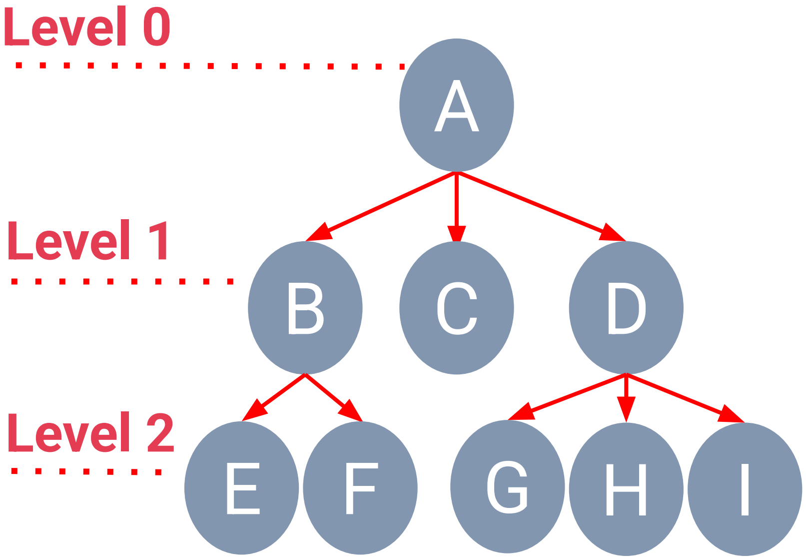
A ***Degree of node*** is a number of edges connected to the node.



- ***Root*** Node: *A*
- ***Parents***: *A*, *B*, and *D*
- ***Children***: *B*, *E*, *F*, *C*, *D*, *G*, *H* and *I*
- ***Siblings***: {*B*, *C*, *D*}, {*E*, *F*}, {*G*, *H*, *I*}
- ***Leaf*** Nodes: *C*, *E*, *F*, *G*, *H* and *I*
- Degree of Tree is 8
- Degree of Node D is 3

Basic Elements of Tree Structure

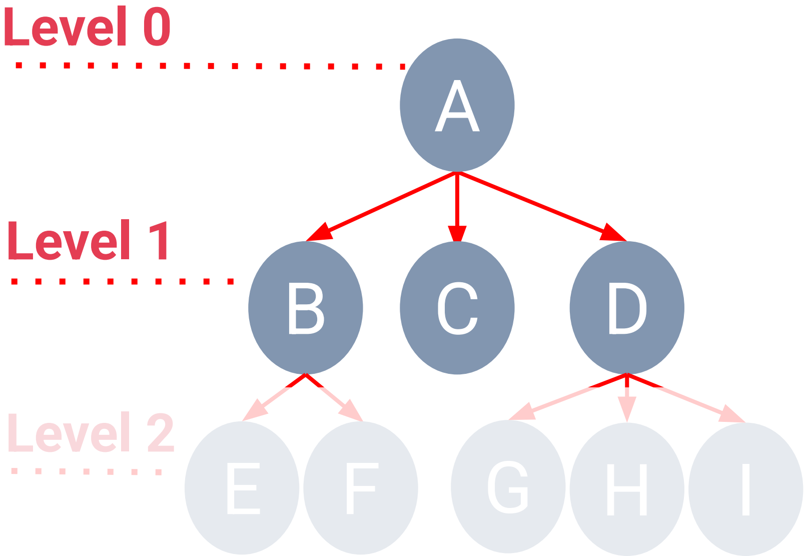
Height of tree is a number of levels starting from the root node.



- ***Root*** Node: *A*
- ***Parents***: *A*, *B*, and *D*
- ***Children***: *B*, *E*, *F*, *C*, *D*, *G*, *H* and *I*
- ***Siblings***: {*B*, *C*, *D*}, {*E*, *F*}, {*G*, *H*, *I*}
- ***Leaf*** Nodes: *C*, *E*, *F*, *G*, *H* and *I*
- Degree of Tree is 8
- Degree of Node D is 3
- Height of tree is 3

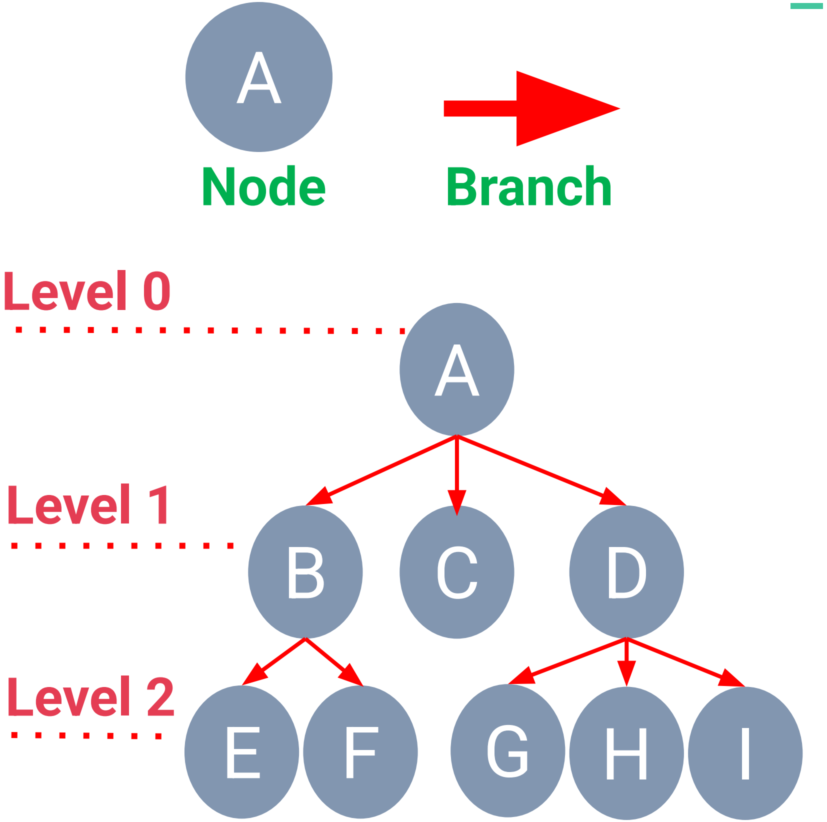
Basic Elements of Tree Structure

Depth of a node is a length of the path to its root.



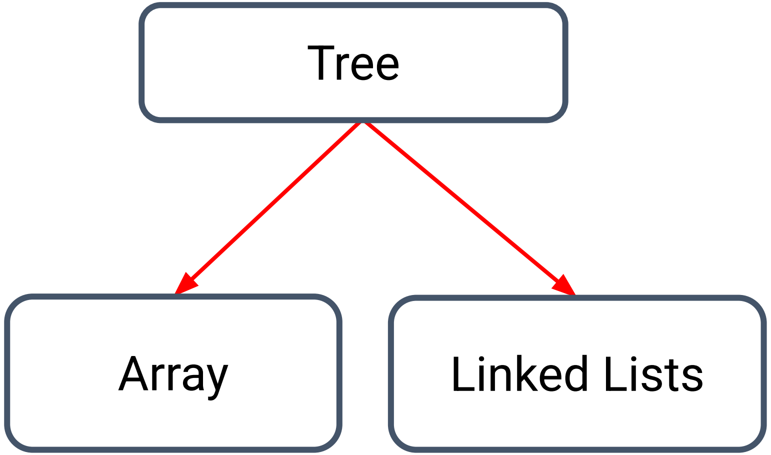
- ***Root*** Node: *A*
- ***Parents***: *A*, *B*, and *D*
- ***Children***: *B*, *E*, *F*, *C*, *D*, *G*, *H* and *I*
- ***Siblings***: {*B*, *C*, *D*}, {*E*, *F*}, {*G*, *H*, *I*}
- ***Leaf*** Nodes: *C*, *E*, *F*, *G*, *H* and *I*
- Degree of Tree is 8
- Degree of Node D is 3
- Height of tree is 3
- Depth of node D is 1

Basic Elements of Tree Structure

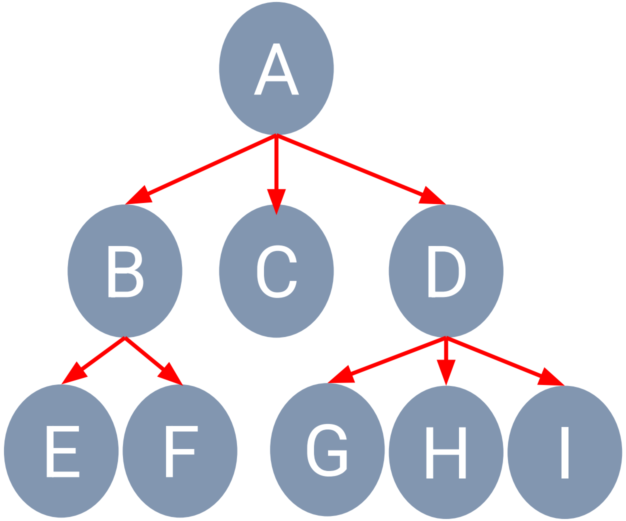


- ***Root*** Node: *A*
- ***Parents***: *A*, *B*, and *D*
- ***Children***: *B*, *E*, *F*, *C*, *D*, *G*, *H* and *I*
- ***Siblings***: {*B*, *C*, *D*}, {*E*, *F*}, {*G*, *H*, *I*}
- ***Leaf*** Nodes: *C*, *E*, *F*, *G*, *H* and *I*
- Degree of Tree is 8
- Degree of Node D is 3
- Height of tree is 3
- Depth of node D is 1

Implementation of Trees



Other Implementation of Trees

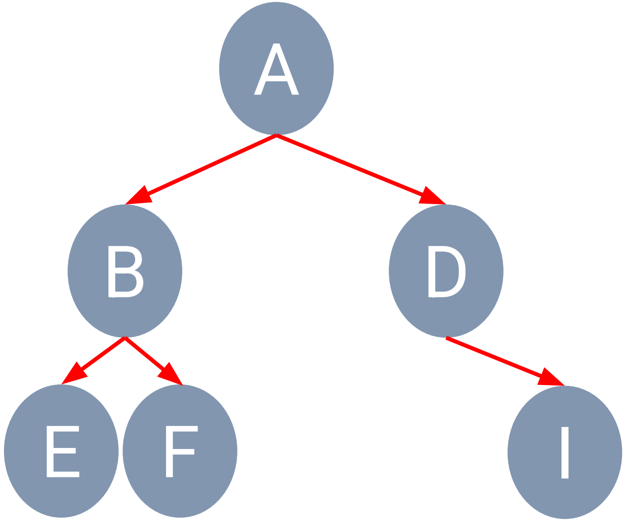


A{ B{E, F}, C, D{G, H, I} }

Linked List Form

Set/List Form

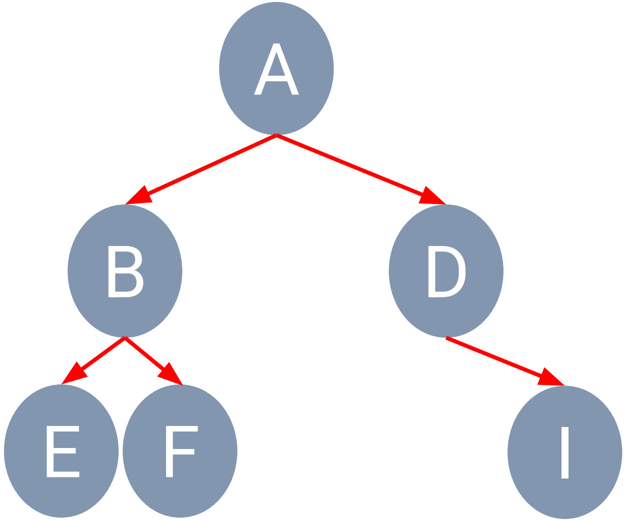
Binary Trees



- A **binary tree** is a tree in which any node can have only two children at maximum.
- Each child node is designated as being either a **left child** or **right child**.

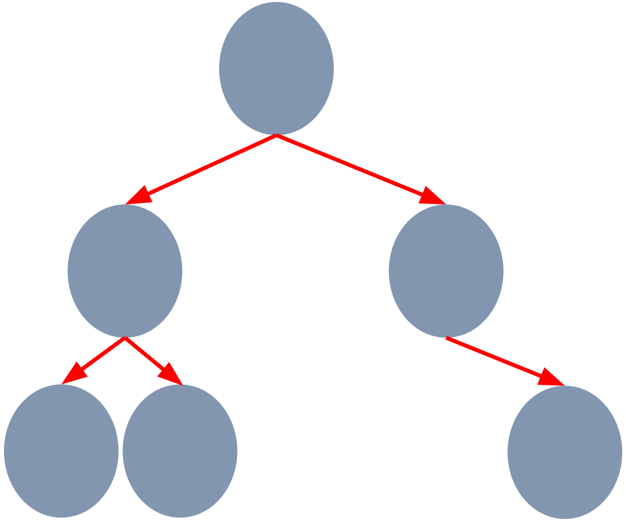
[1] Michael T. Goodrich et al., Data Structures and Algorithms in Python, 2013

Recursive Binary Trees



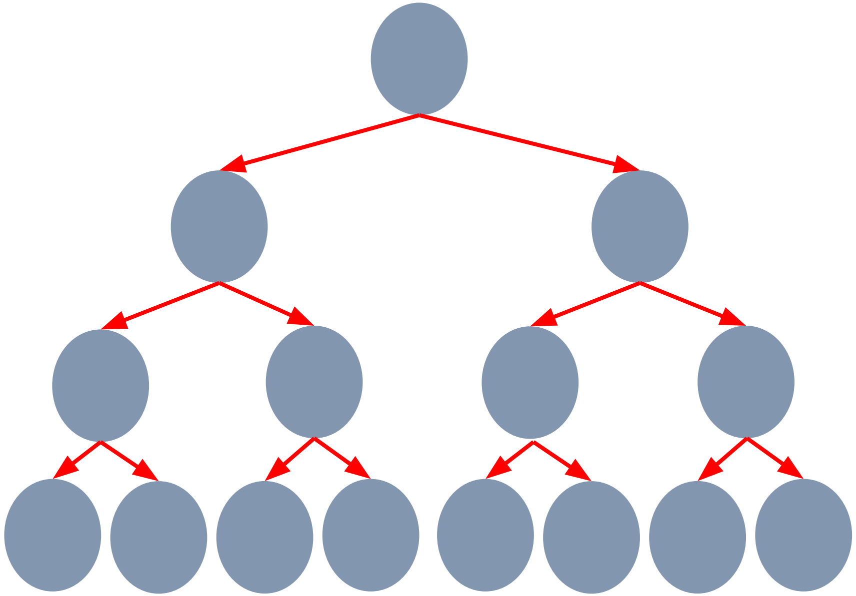
- A **binary tree** is either empty or consists of:
  - A root node that stores an element
  - A binary left subtree (possibly empty)
  - A binary right subtree (possibly empty)

Binary Trees' Properties



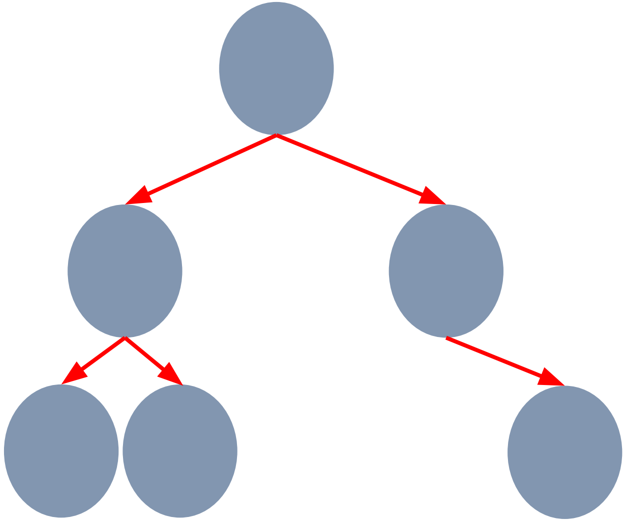
- A **Degree** of node is a number of edges/subtrees connected to the node.
- A **Depth** of a node is a length of the path to its root (i.e. number of edges from root the node).
    - d denotes the depth/level of a node
- **Height** of tree is a number of levels starting from the root node (i.e. number of edges from the node to the deepest leaf).
    - *h* denotes the height of tree

Binary Trees’ Properties



		Number of Nodes In each leavel
Level 0		**1**
Level 1		**2**
Level 2		**4**
Level 3		**8**
Level d		**2^d**

Binary Trees' Properties



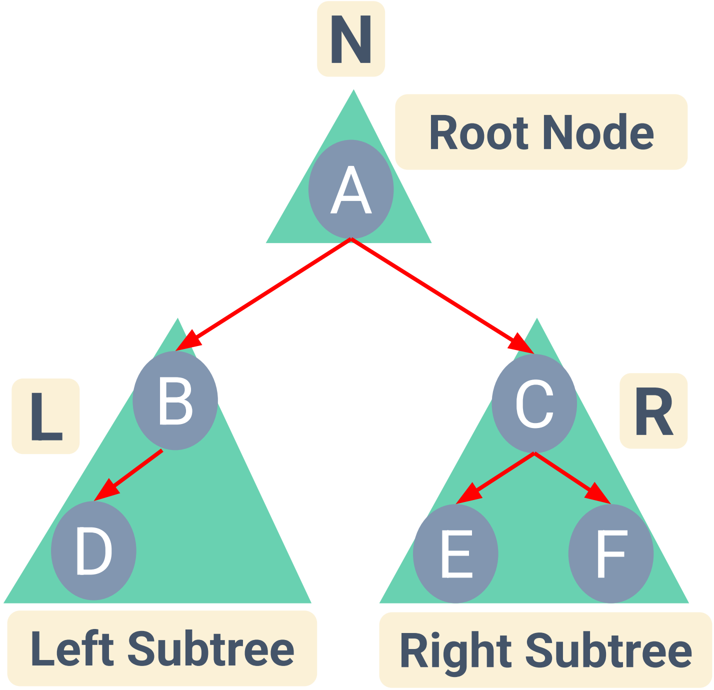
Suppose n is the total number of nodes in a tree and h is the number of height of tree.

- **Maximum height** of binary trees, h_{max} =
- **Minimum height** of binary trees, h_{min} =
- **Minimum number** of nodes in a tree, n_{min} =
- **Maximum number** of nodes in a tree, n_{max} =

Binary Tree Traversal

- A **traversal** of a tree is a method to access all the tree nodes.
- Two main approaches: Depth-first and Breadth-first
  - Depth-first
    - Preorder traversal
    - Postorder traversal
    - Inorder traversal
  - Breadth-first - visit all the nodes at depth *d* before moving to depth *d*+1.

Depth-first Traversal

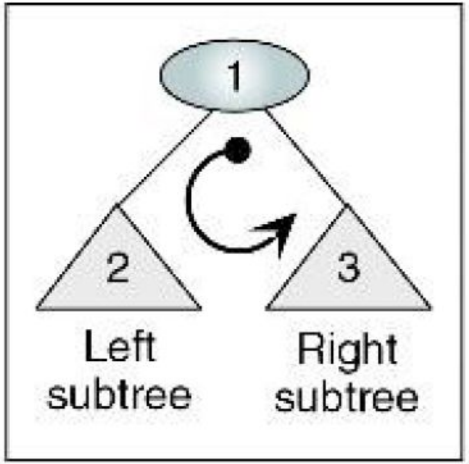


- **Pre**order Traversal
  N L R
- **In**order Traversal
  L N R
- **Post**order Traversal
  L R N

Preorder Traversal

- Root -> Left subtree -> Right subtree

```
Algorithm preOrder(root):
    if (root is not null):
        process(root)
        preOrder(leftSubtree)
        preOrder(rightSubtree)
    end if
end preOrder
```

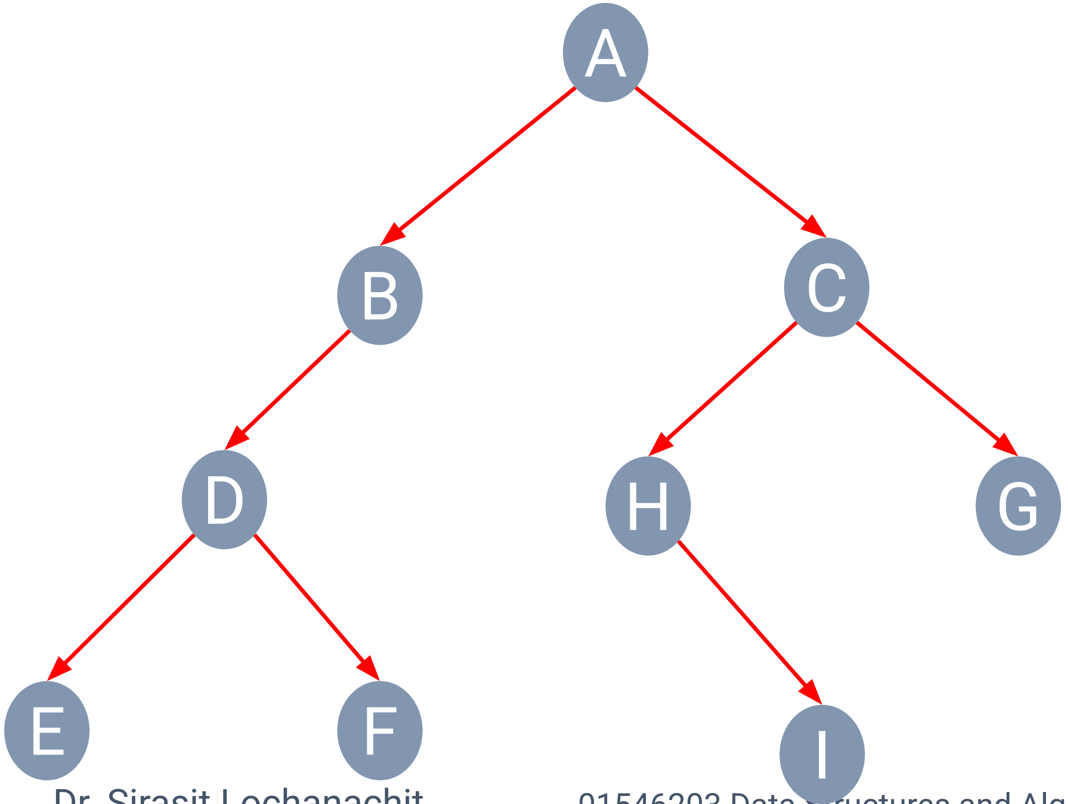


Preorder Traversal

- **Preorder Traversal**

N L R

- **Visited nodes**



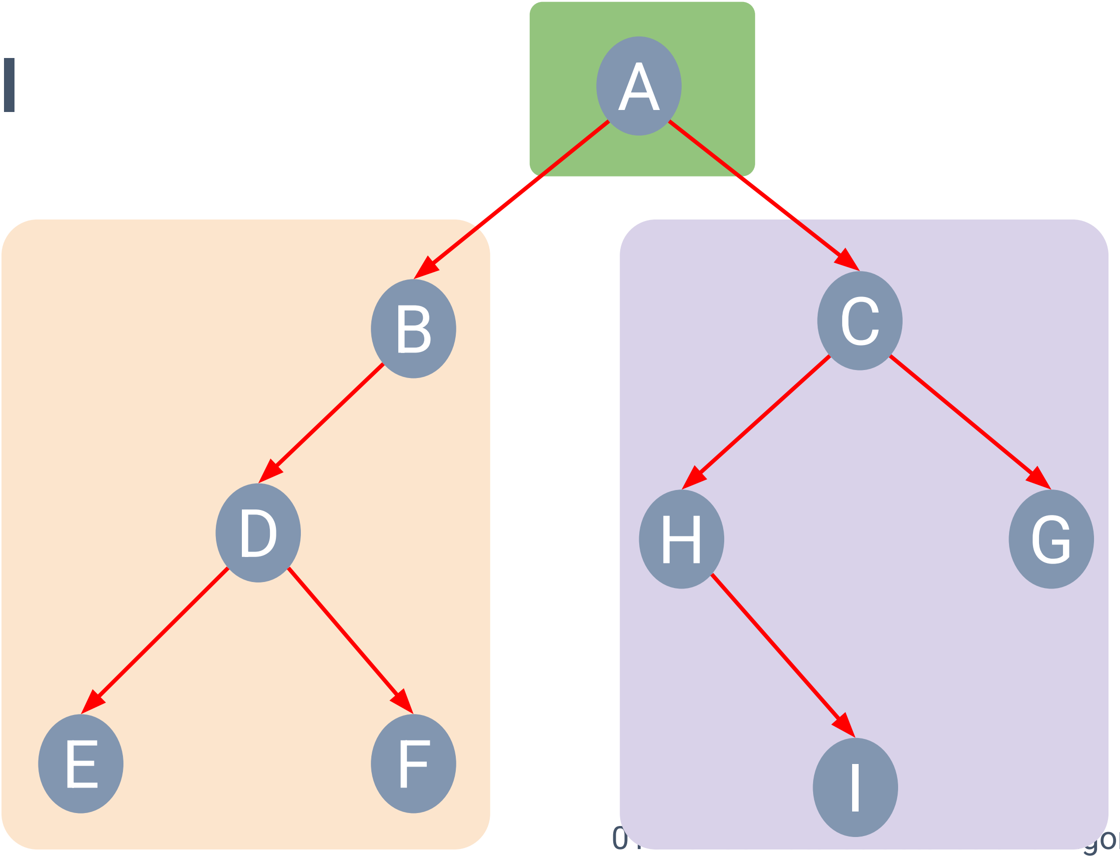
Preorder Traversal

- **Preorder Traversal**

N L R

- **Visited nodes**

A



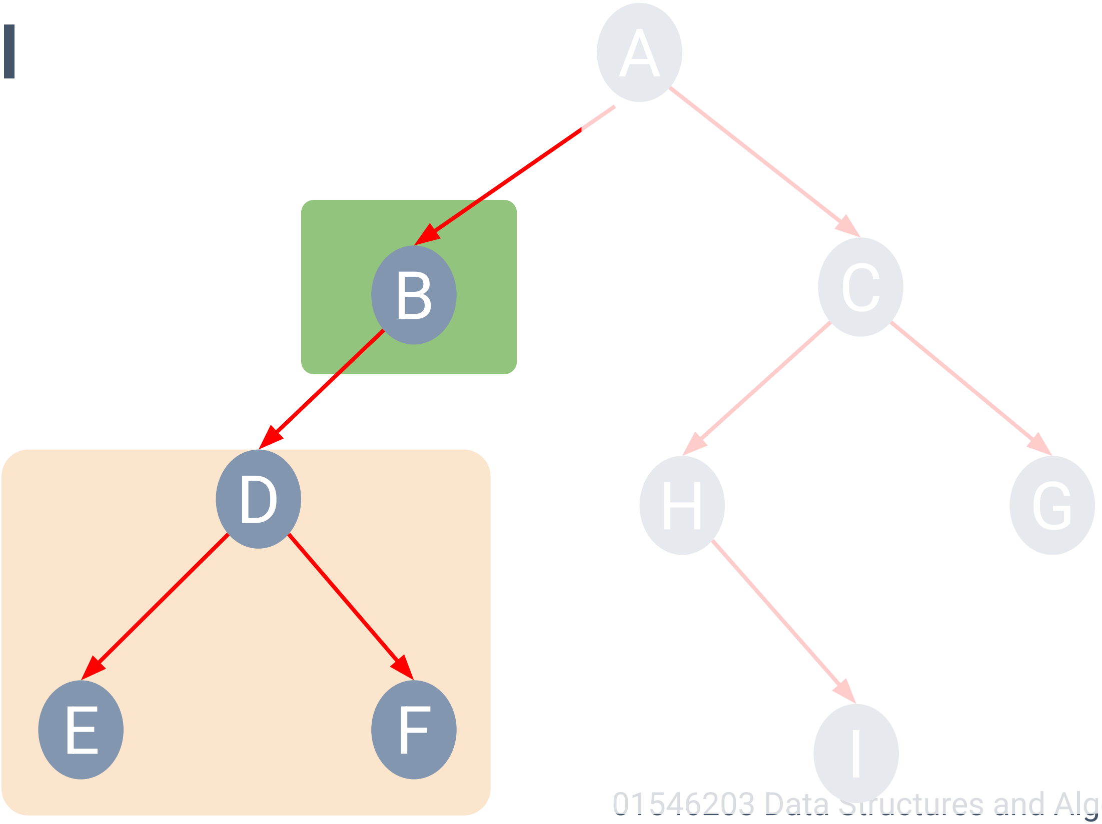
Preorder Traversal

- **Preorder Traversal**

N L R

- **Visited nodes**

A B



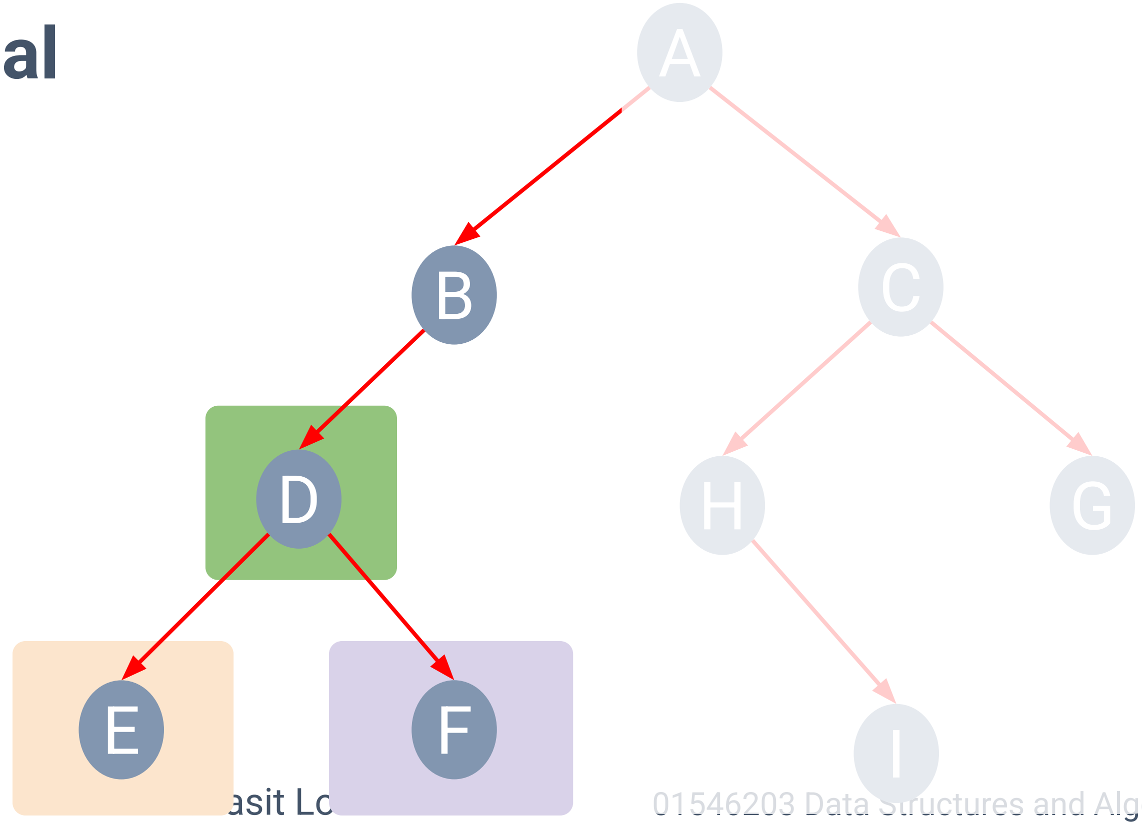
Preorder Traversal

- **Preorder Traversal**

N L R

- **Visited nodes**

A B D E F



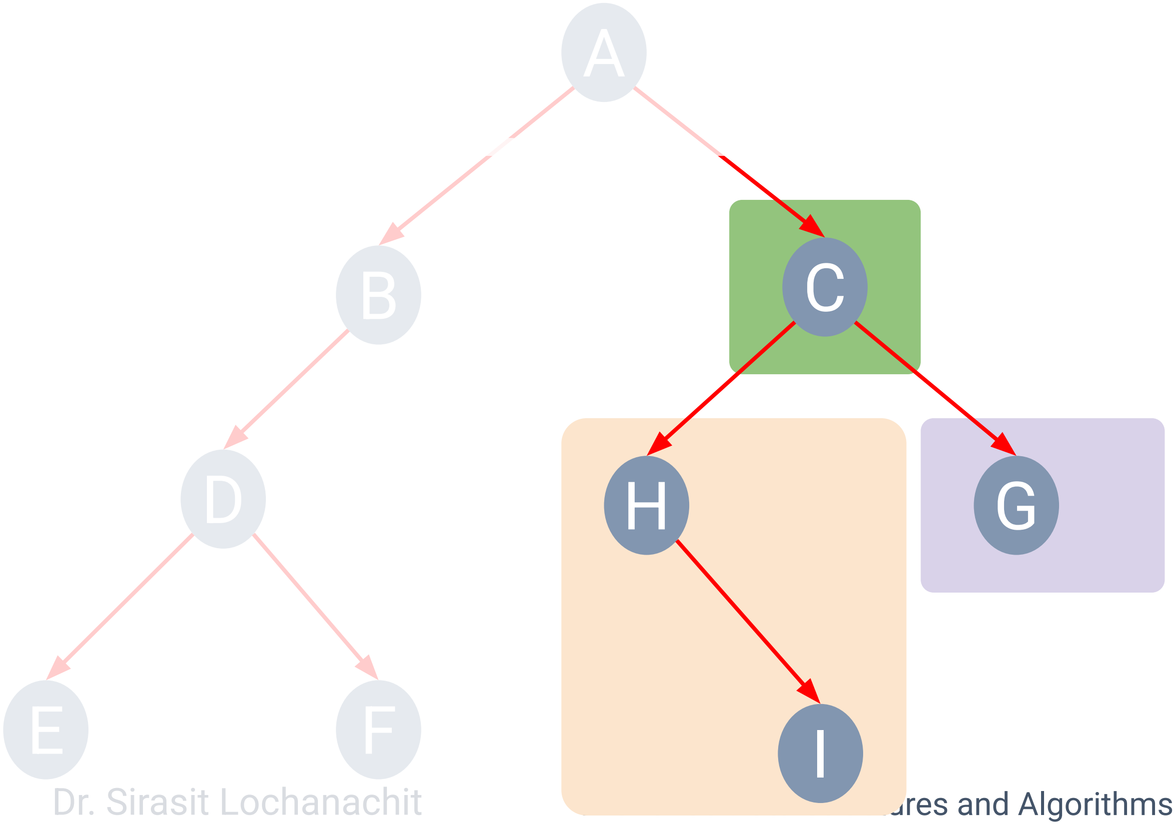
Preorder Traversal

- **Preorder Traversal**

  N L R

- **Visited nodes**

A B D E F



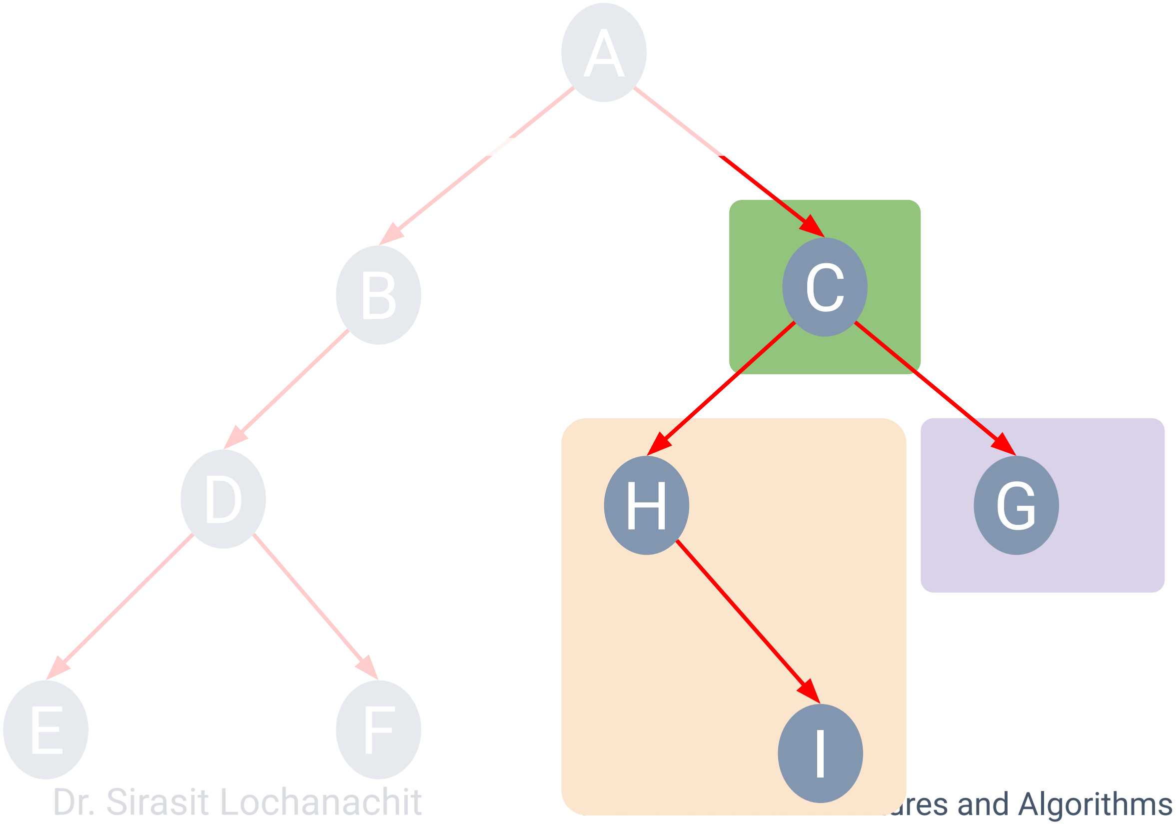
Preorder Traversal

- **Preorder Traversal**

N L R

- **Visited nodes**

A B D E F C



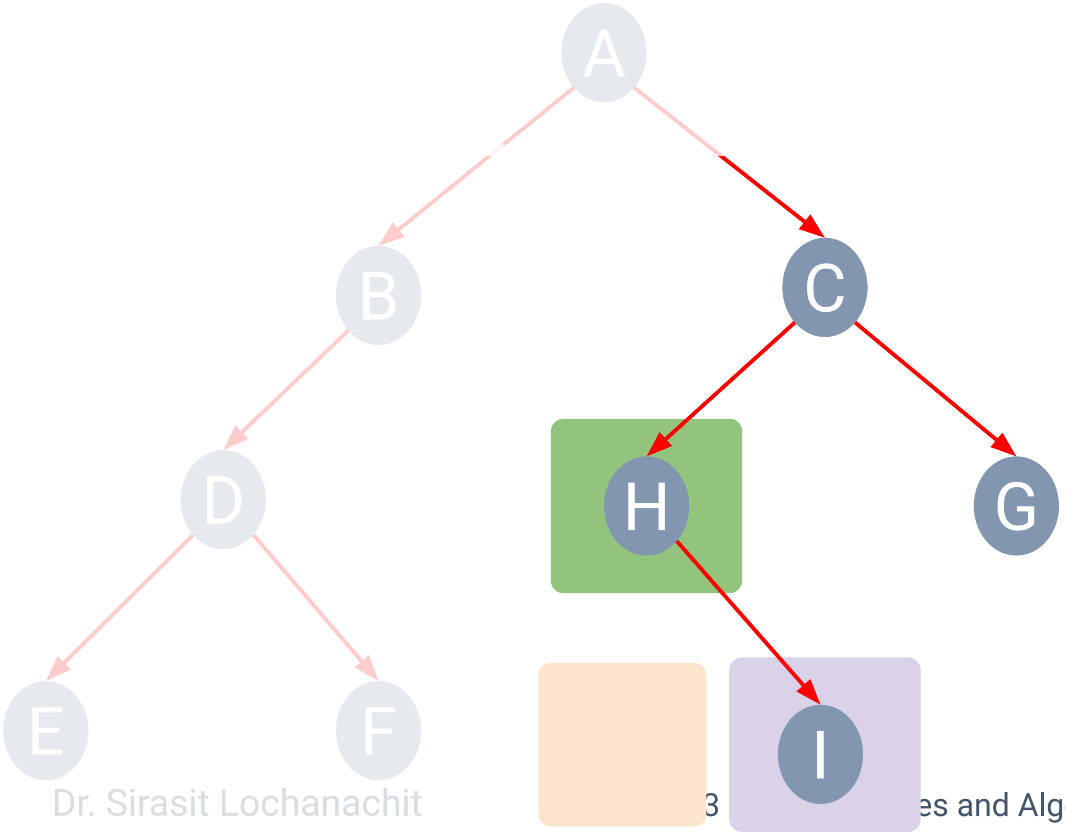
Preorder Traversal

- **Preorder Traversal**

N L R

- **Visited nodes**

A B D E F
C H I



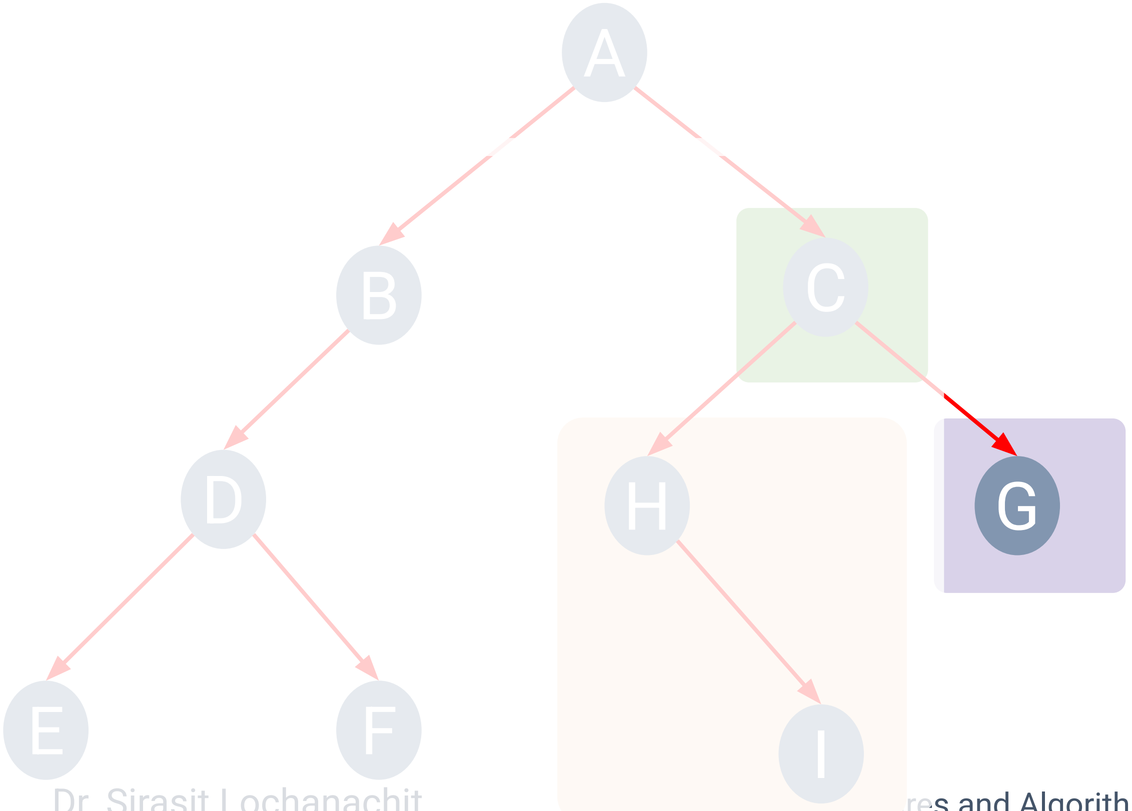
Preorder Traversal

- **Preorder Traversal**

N L R

- **Visited nodes**

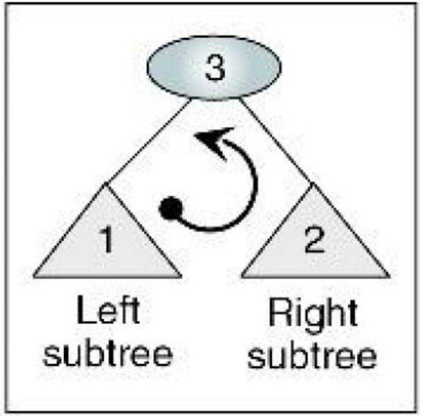
A B D E F
C H I G



Postorder Traversal

- Left subtree -> Right subtree -> root

```
Algorithm postOrder(root):
    if (root is not null):
        postOrder(leftSubTree)
        postOrder(rightSubtree)
        process(root)
    end if
end postOrder
```

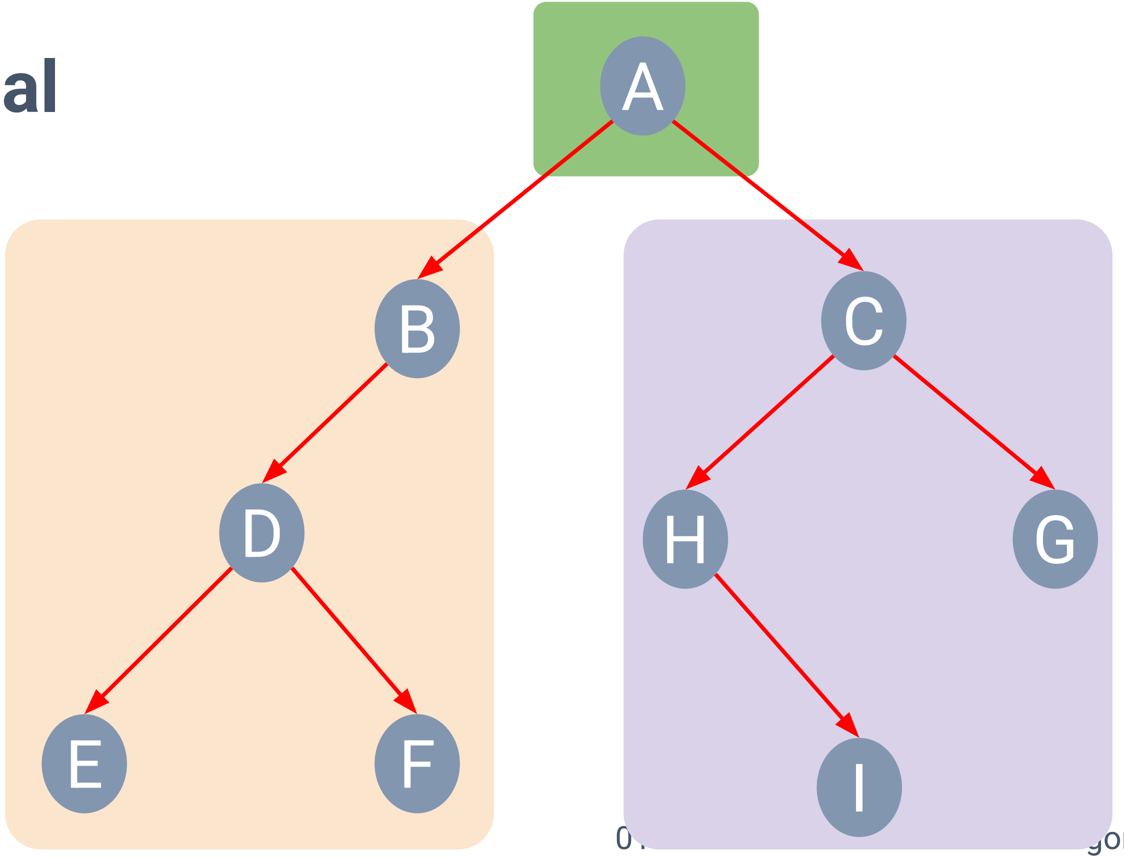


Postorder Traversal

- **Postorder Traversal**

  L R N

- **Visited nodes**

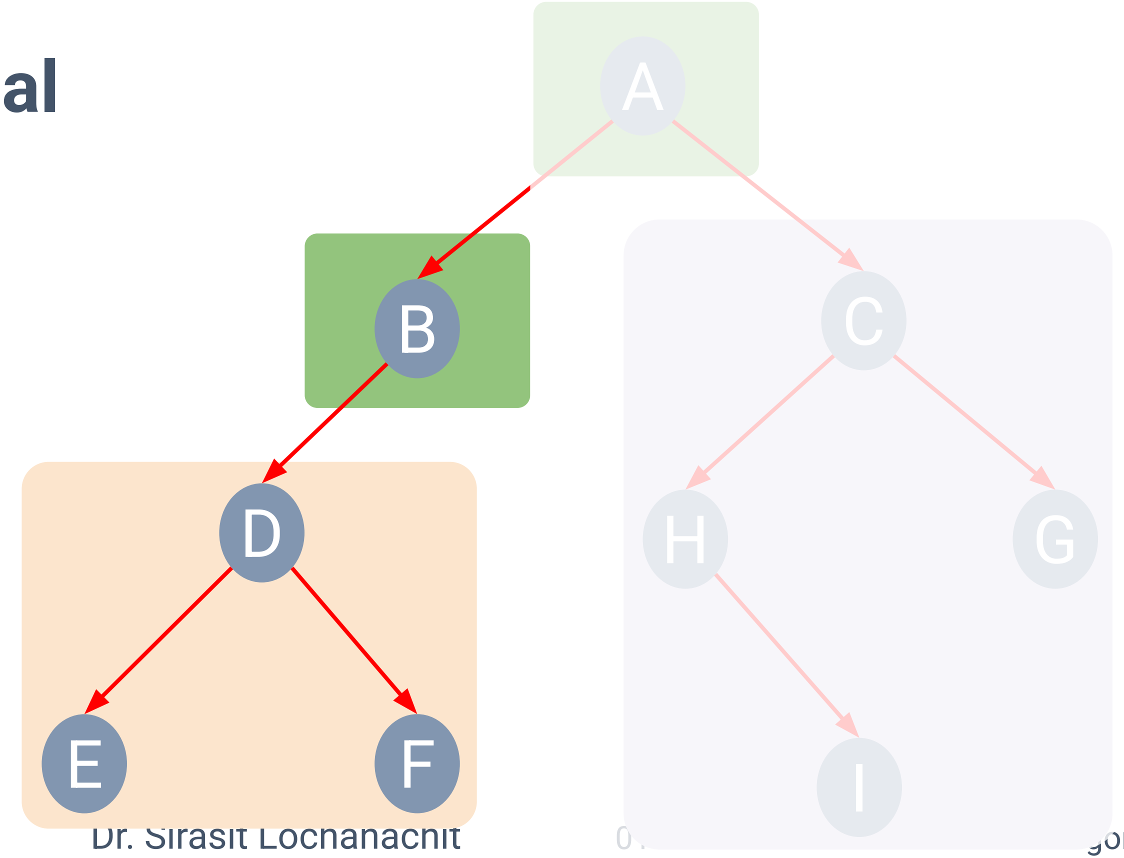


Postorder Traversal

- **Postorder Traversal**

L R N

- **Visited nodes**



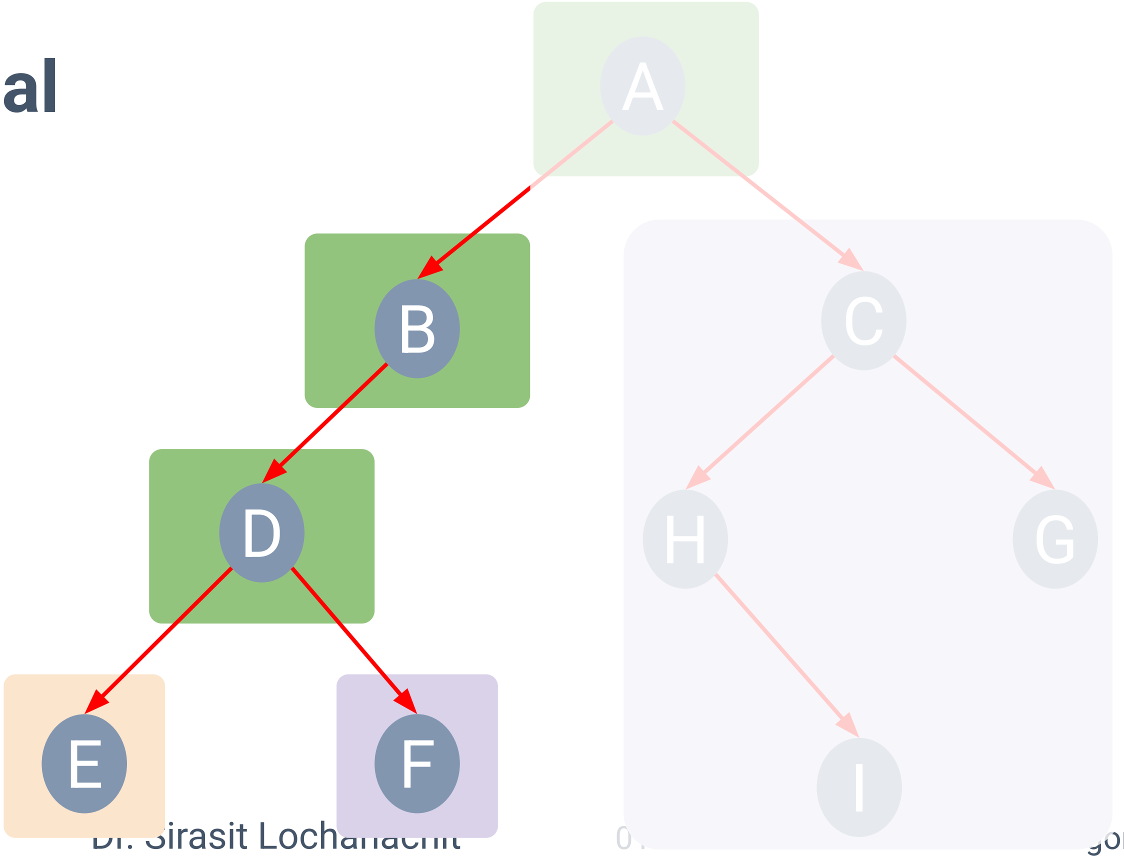
Postorder Traversal

- **Postorder Traversal**

L R N

- **Visited nodes**

E F



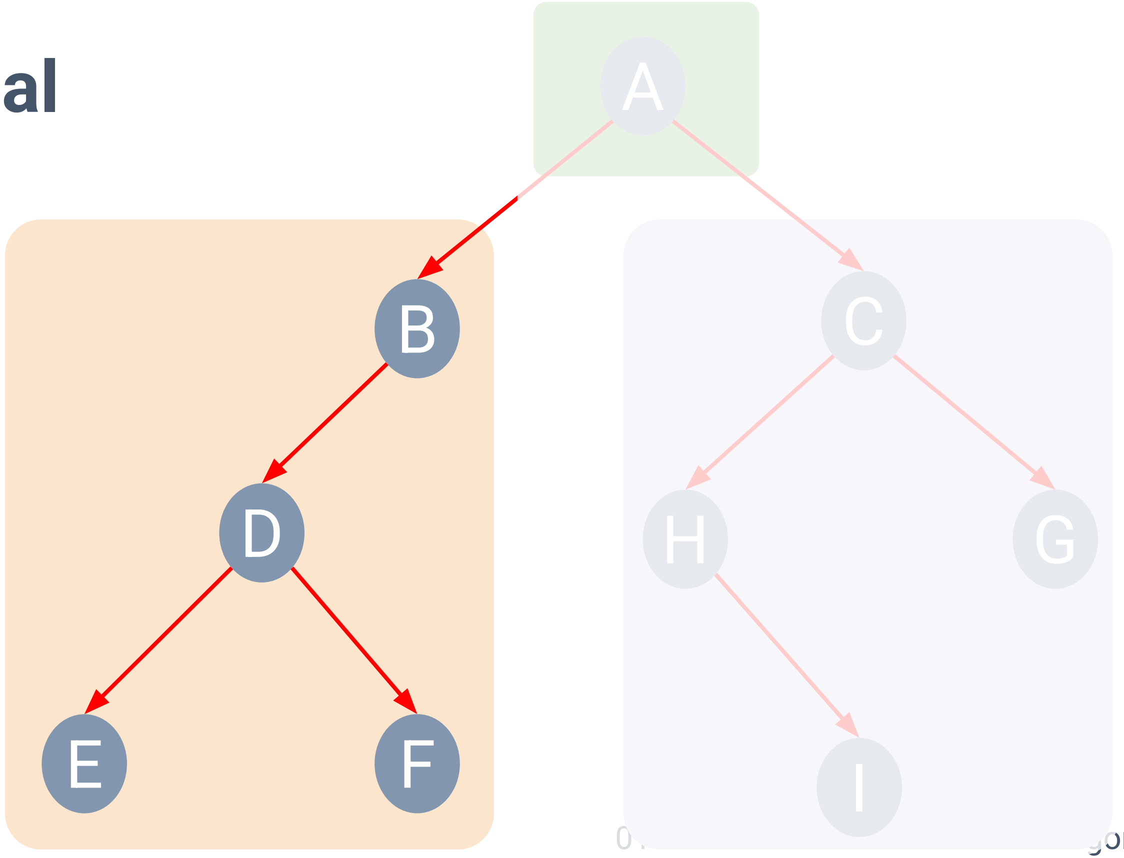
Postorder Traversal

- **Postorder Traversal**

  L R N

- **Visited nodes**

E F D B



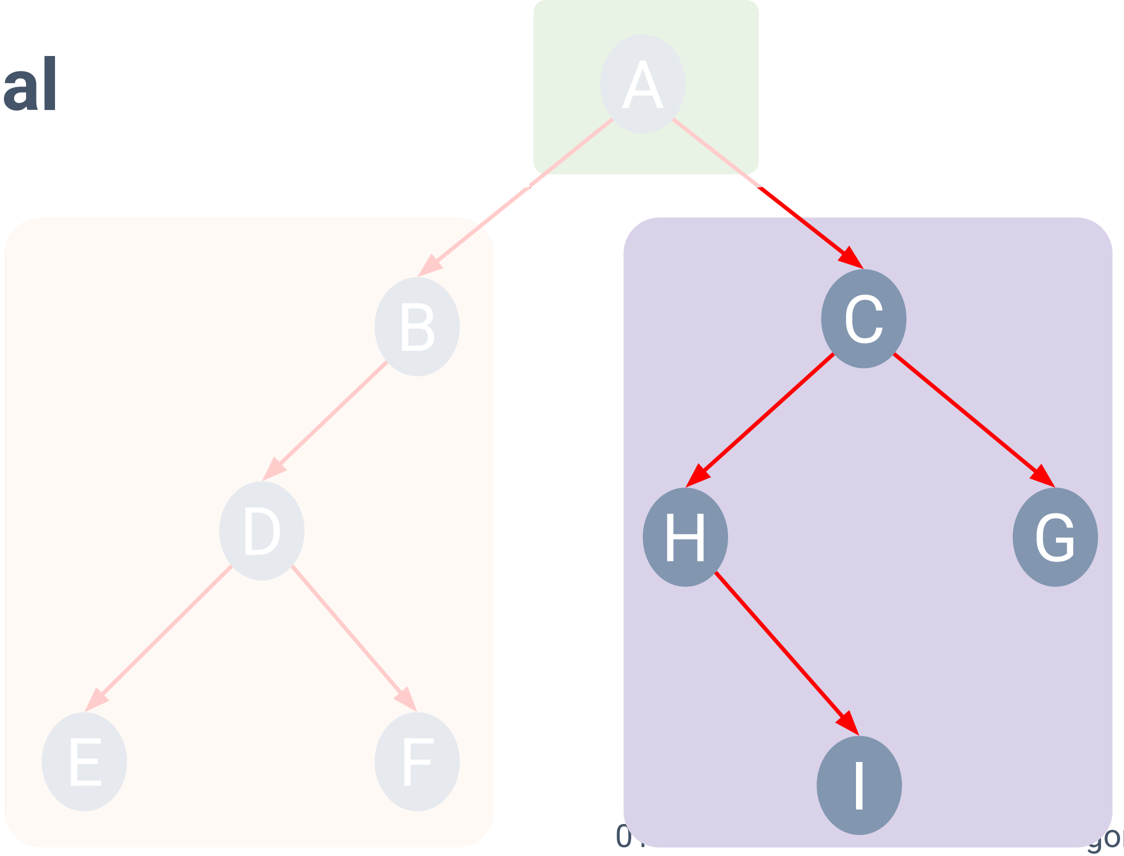
Postorder Traversal

- **Postorder Traversal**

L R N

- **Visited nodes**

E F D B
I H G C



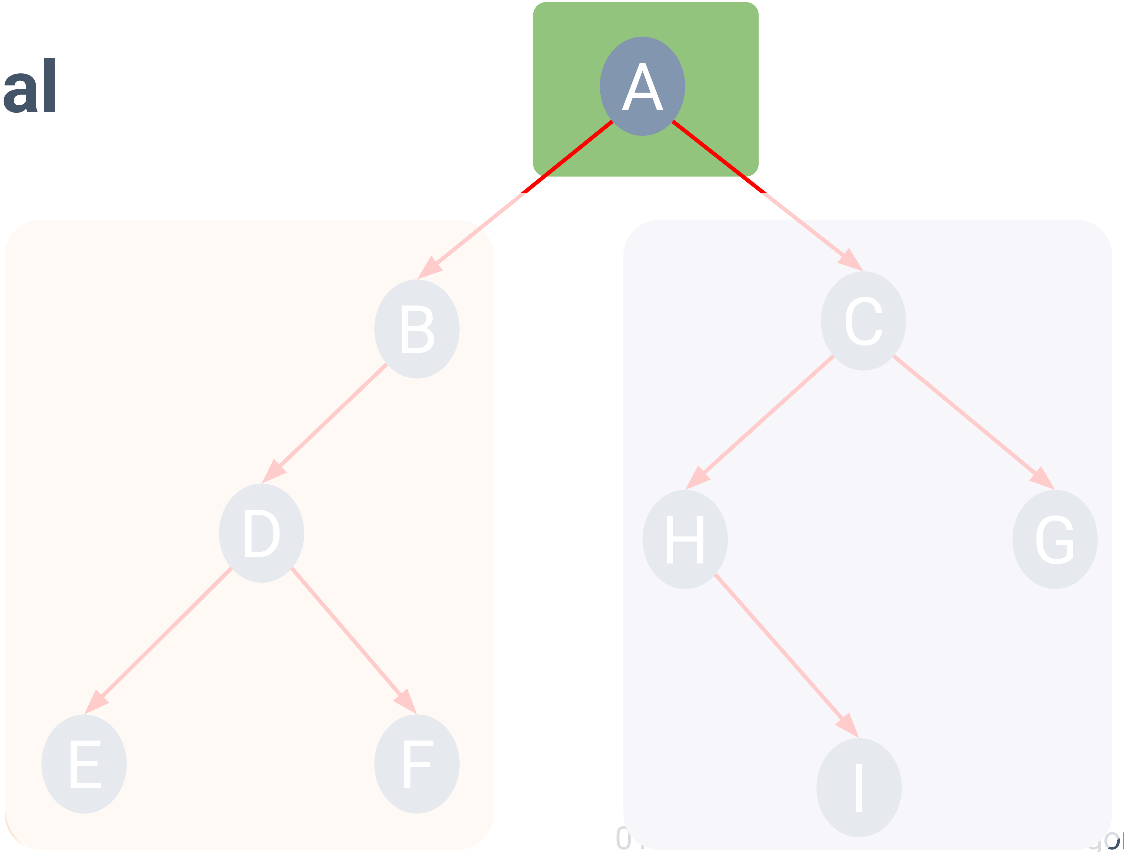
Postorder Traversal

- **Postorder Traversal**

L R N

- **Visited nodes**

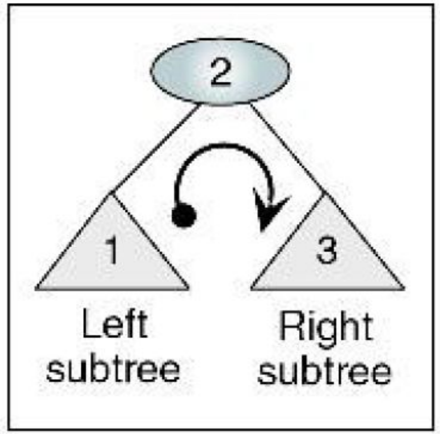
E F D B
I H G C
A



Inorder Traversal

- Left subtree -> root -> Right subtree

```
Algorithm inOrder(root):
    if (root is not null):
        inOrder(leftSubTree)
        process(root)
        inOrder(rightSubtree)
    end if
end inOrder
```

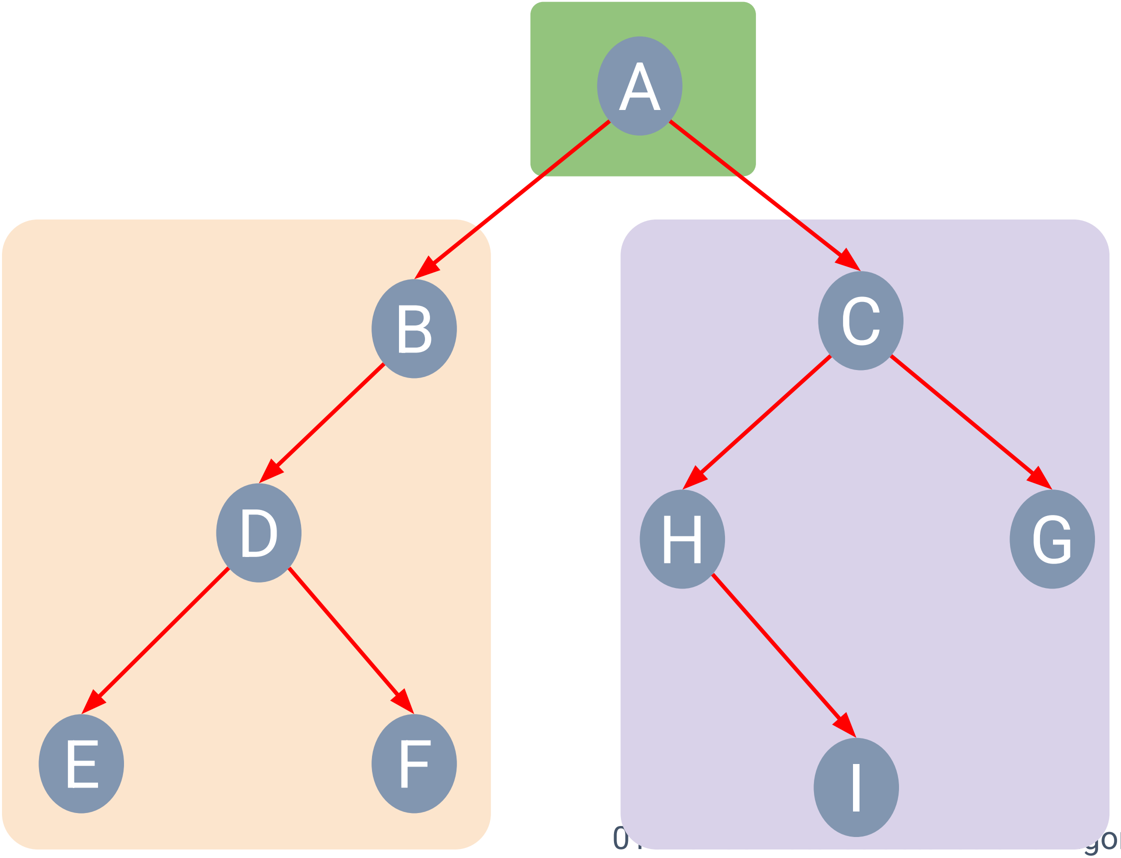


Inorder Traversal

- **Inorder Traversal**

  L N R

- **Visited nodes**



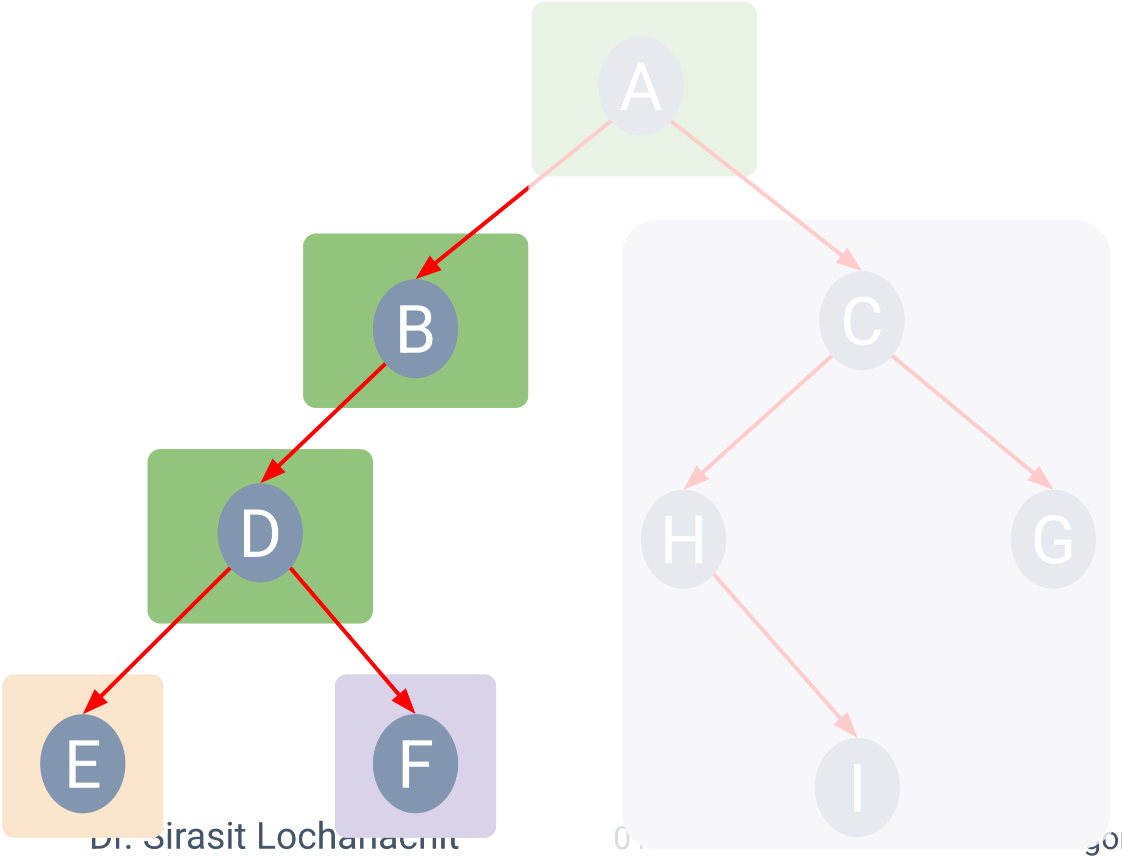
Inorder Traversal

- **Inorder Traversal**

  L N R

- **Visited nodes**

  E D F



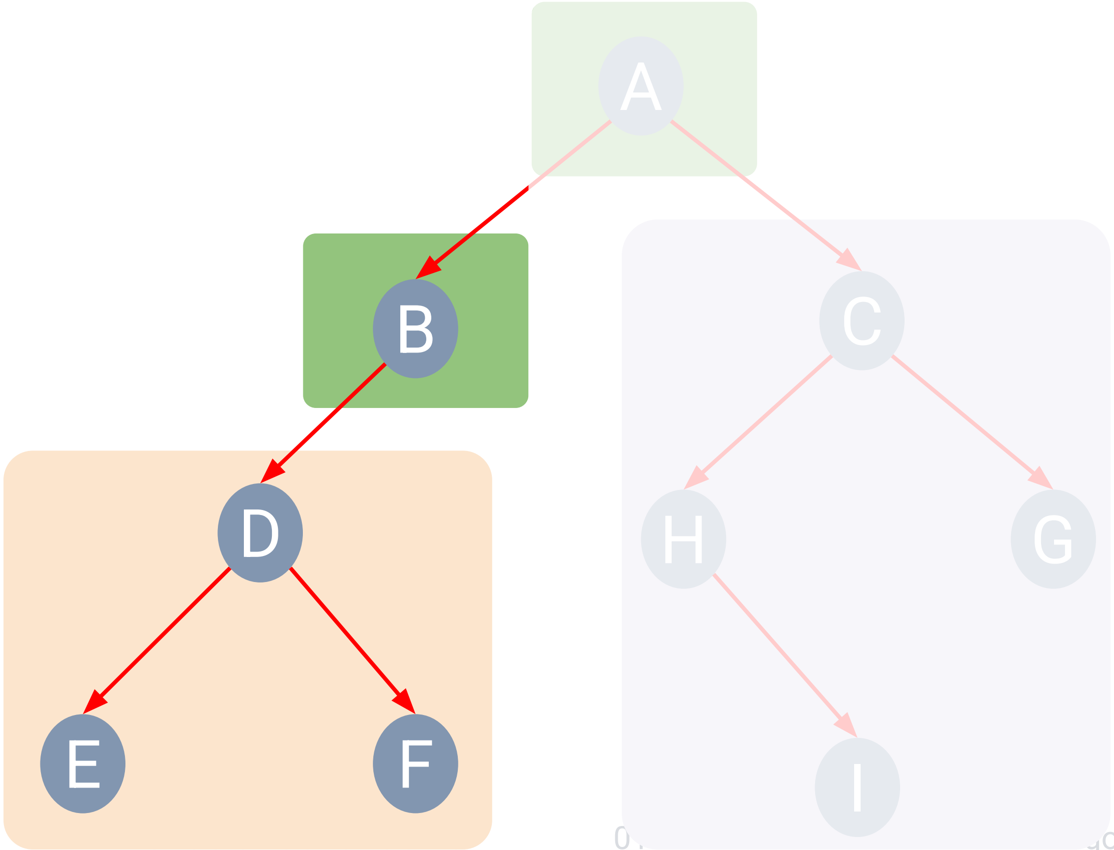
Inorder Traversal

- **Inorder Traversal**

  L N R

- **Visited nodes**

  E D F B



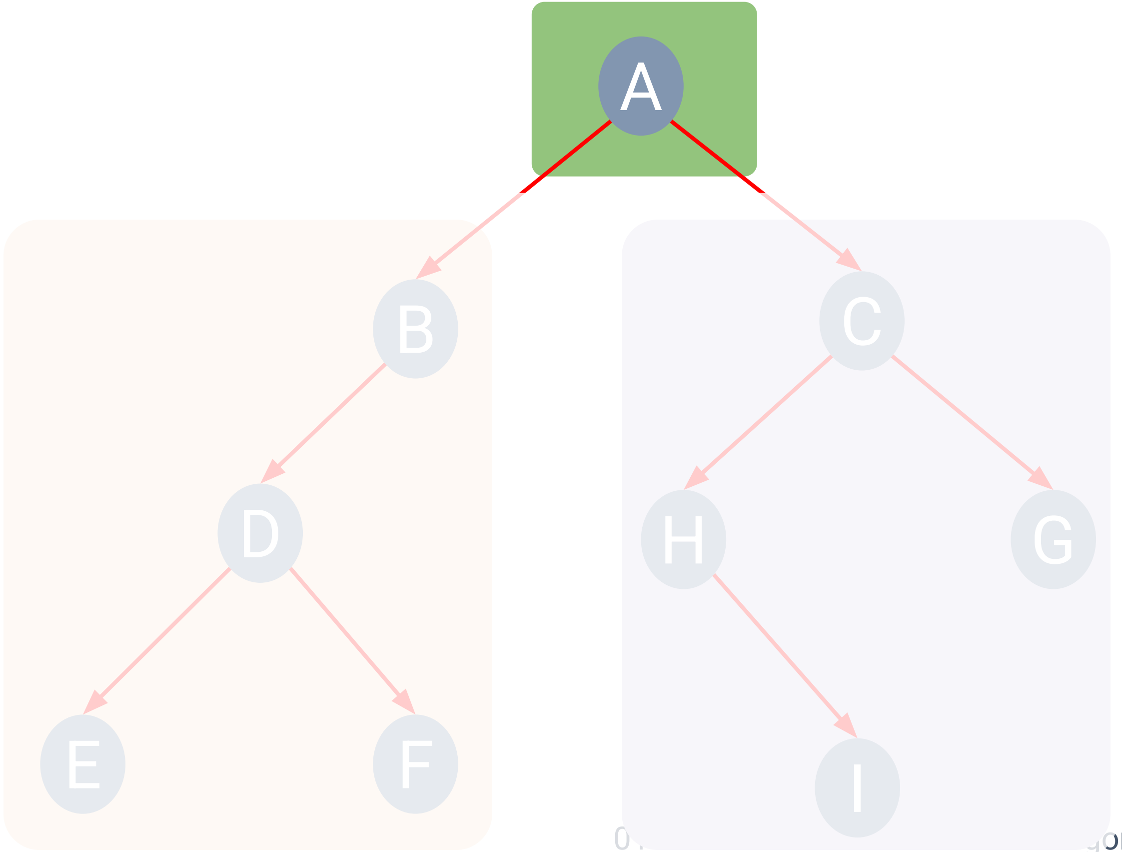
Inorder Traversal

- **Inorder Traversal**

L N R

- **Visited nodes**

E D F B A



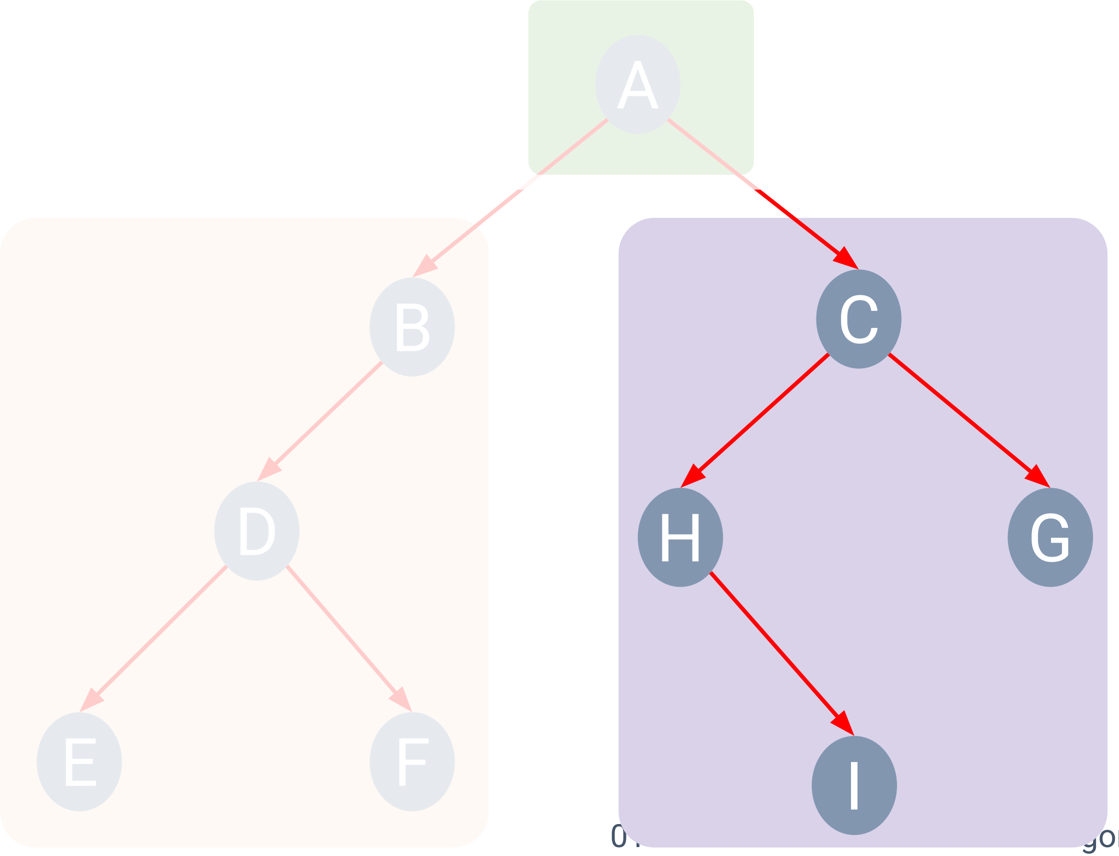
Inorder Traversal

- **Inorder Traversal**

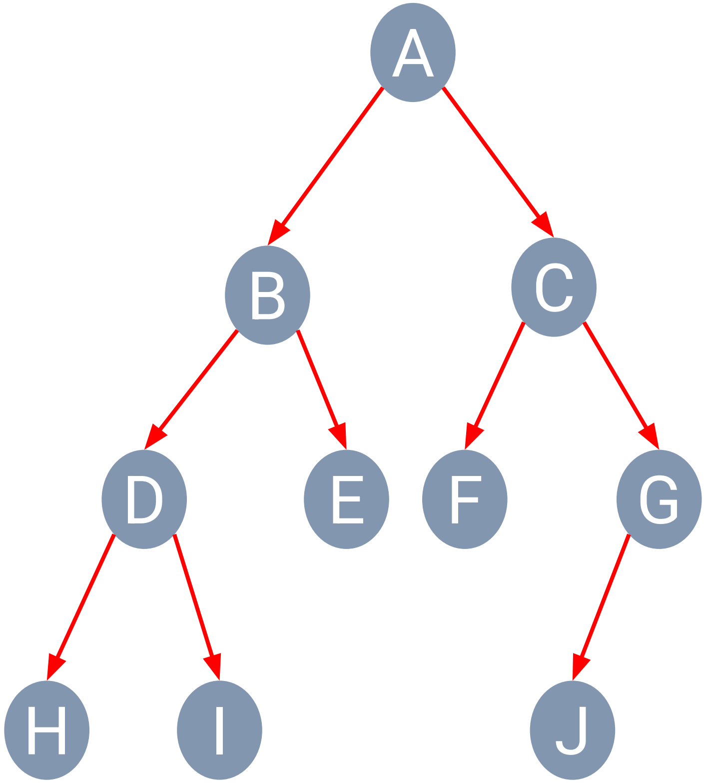
  L N R

- **Visited nodes**

  E D F B A
  H I C G

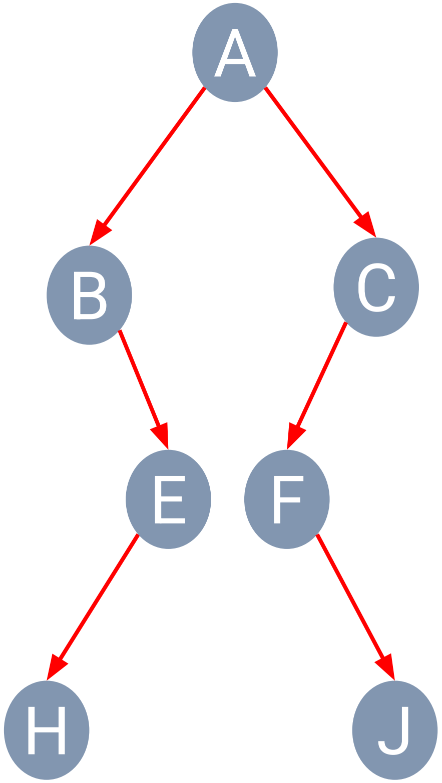


Depth-first Traversal Exercise 1



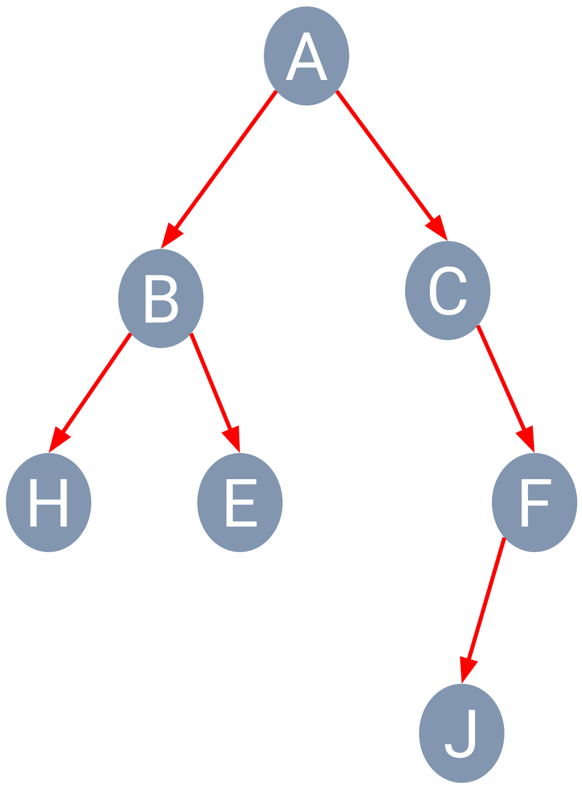
- **Preorder Traversal** N L R
- **Inorder Traversal** L N R
- **Postorder Traversal** L R N

Depth-first Traversal Exercise 2



- Preorder Traversal N L R
- Inorder Traversal L N R
- Postorder Traversal L R N

Depth-first Traversal Exercise 3



- **Pre**order Traversal N L R
- **In**order Traversal L N R
- **Post**order Traversal L R N

Quiz

1. Find root of Binary Tree from a given traversal path:
   a. Tree with postorder traversal: FCBDG
   b. Tree with preorder traversal: IBCDFEN
   c. Tree with inorder traversal: CBIDFGE

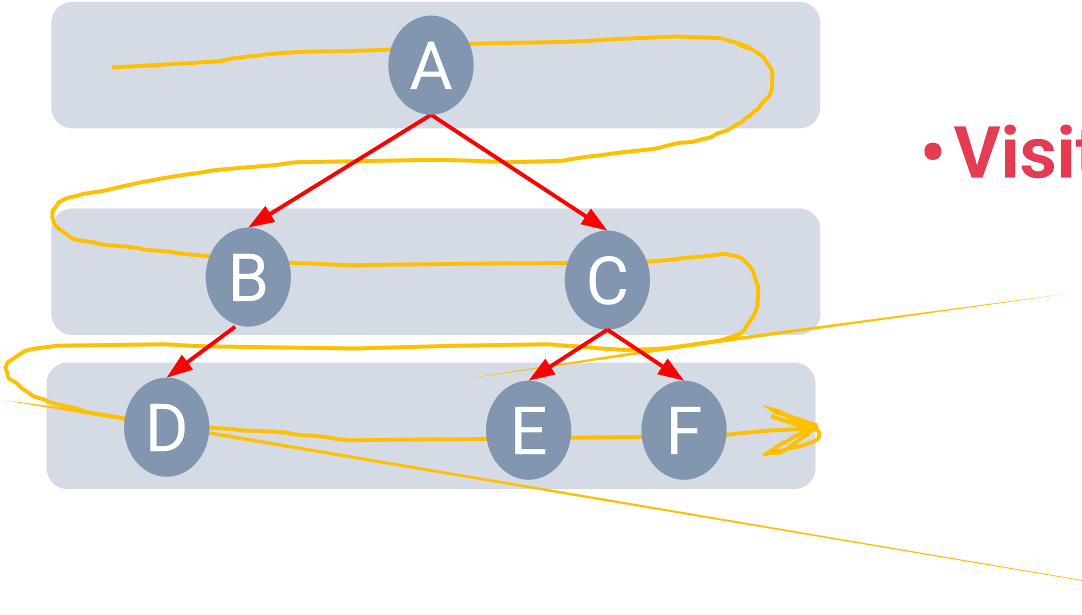
Quiz

2. Draw a Binary Tree from a given traversal path:

    a. Preorder: JCBADEFIGH

    b. Inorder: ABCEDFJGIH

Breadth-first Traversal



- Visited nodes

- Visit all the nodes at depth *d* before moving to depth *d*+1.